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MANAGEMENT OF AIR FORCE OPERATIONAL TRAINING AND UNDERGRADUATE AIRCREW TRAINING SYSTEMS

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This publication implements Department of the Air Force Policy Directive (DAFPD) 10-2, *Readiness*. It provides guidance and procedures for managing, developing, and fielding operational training and undergraduate aircrew training systems. This publication applies to all civilian employees and uniformed members of the Regular Air Force, the Air Force Reserve Command (AFRC), the Air National Guard (ANG), and those with a contractual obligation to abide by the terms of United States Air Force (USAF) issuances. This instruction does not apply to the United States Space Force (USSF). Conflicts between this instruction and Joint Program Office (JPO) direction will be reconciled by the F-35 Integration Office (AF/IO), Air Combat Command (Lead Command), and the JPO. The authorities to waive wing, unit, delta, or garrison level requirements in this publication are identified with a Tier number ("T-0, T-1, T- 2, T-3") following the compliance statement. Submit requests for waivers through the chain of command to the

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(ACC) AFI 10-217, *Management of Air Force Operational Training and Undergraduate Aircrew Training Systems*, is supplemented as follows: This supplement prescribes guidance and procedures governing Operational Training System (TS) operation and management for Air Combat Command (ACC). This publication applies to all ACC -assigned and attached military and civilian Air Force (AF) personnel, members of the Air Force Reserve Command (AFRC), National Guard Bureau (ANG), and contractor support personnel in accordance with (IAW) appropriate provisions contained in memoranda support agreements and AF contracts. This publication does not apply to the United States Space Force. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Send comments and suggested improvements to this supplement on Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*, through channels to ACC/A3T Flight Operations Division Org acc.A3T@us.af.mil, and ACC/A3TO Flight Operations and Training Branch Org, acc.a3to@us.af.mil, or ACC Flight Operations (ACC/A3T) 205 Dodd Blvd, Ste 121, Langley AFB VA 23665. This publication may be supplemented at any level, but all Supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. The authorities to waive wing/unit level requirements in this publication are identified with a tier ("T-2 or T-3") number following the compliance statement. See Department of the Air Force (DAFMAN) 90-161, *Publishing Processes and Procedures*, Table A10.1, for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority IAW DAFMAN 90-161, Chapter 9. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been revised and needs to be completely reviewed. It supersedes AFI16-1007; 1 October 2019. This revision reflects the activation of USSF as a service and removes the space mission from the USAF; standardizes USAF to denote the Air Force service IAW Department of the Air Force Manual (DAFMAN) 90-161, *Publications Processes and Procedures*. Tiering and compliance statements are IAW Administrative Assistant to the Secretary of the Air Force recommendations. It includes new or modified guidance on operational training, operational

training and undergraduate aircrew training systems, training system responsibilities and management, and reporting on training systems. Changes include a category for Synthetic Environment certification, institutes standards and criteria for capturing simulator utilization data (see **attachment 4**), and includes unclassified threat matrix framework (TMF) information (see **attachment 6**).

(ACC) This publication has been revised and needs to be completely reviewed. It replaces AFI16-1007. This revision reflects the activation of USSF as a service and removes the space mission from the USAF; standardizes USAF to denote the Air Force service IAW DAFMAN 90-161. Tiering and compliance statements are IAW Administrative Assistant to the Secretary of the Air Force (SAF/AA) recommendations. It includes new or modified guidance on operational training systems, and undergraduate aircrew TS, TS responsibilities and management, and reporting on TS. Changes include a category for Synthetic Environment certification, institution of standards and criteria for capturing simulator utilization data (see **Attachment 4**), and includes unclassified TMF information (see **Attachment 6**).

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Chapter 1

GENERAL INFORMATION

1.1. Purpose. This instruction provides guidance, identifies organizational responsibilities, and describes processes for managing USAF operational training systems and undergraduate aircrew training systems.

1.1. (ACC) Purpose. This supplement specifies the responsibilities of ACC in supporting the Combat Air Forces (CAF) to operate, manage, and evaluate TSs, Training System Support Centers (TSSC), associated support equipment, courseware, and instruction. The CAF includes the following agencies: Headquarters (HQ) Air Combat Command (ACC), HQ United States Air Forces in Europe – Air Forces Africa, HQ Air Force Global Strike Command, HQ Pacific Air Forces, NGB, and HQ AFRC.

1.2. Scope. The following operational training systems and associated training devices are covered by this AFI:

1.2.1. Aircrew training systems: Includes aeromedical evacuation, physiological training, and undergraduate aircrew training systems.

1.2.2. Missile procedures training systems: Includes missile procedures trainers, minuteman enhanced procedures trainers, airborne procedures trainers, and countdown procedures trainers.

1.2.3. Cyberspace training systems.

1.2.4. Mission system training systems: Includes Joint Terminal Attack Controller (JTAC), Air Support Operations Center, Control and Reporting Center, Air Operations Center (AOC), Air Traffic Control, and those systems which are not considered aircrew training, Cyberspace training, or missile procedures training systems.

1.3. Operational Training Concepts.

1.3.1. Live (real people operating real weapon systems), virtual (real people operating simulated systems), and constructive (computer generated entities) are foundational domains for operational training. Live, synthetic, and blended training environments are defined by their use of live, virtual, and/or constructive capabilities. Live training environments can include either virtual or constructive assets. Synthetic training environments include virtual and/or constructive capabilities, and may be in a networked or non-networked environment. When live and synthetic environments are combined, the training environment is blended.

1.3.2. The Operational Training Infrastructure (OTI) framework tailors a balanced mix of capabilities to create the most affordable and effective training environments for warfighters with the intent of maximizing readiness. The OTI framework includes live ranges, threat emitters, aggressors, networks, training centers, and multi-domain command and control training systems.

1.3.3. Effective operational training includes an efficient balance of live and synthetic approaches. Synthetic training solutions should be used to replace or augment live training to the maximum extent practicable where training effectiveness and readiness are not

compromised, recognizing that some live training events cannot or should not be replaced by synthetic training.

1.3.4. The Operational Training and Test Infrastructure (OTTI) framework consists of the resources essential to accomplishing USAF operational training and testing (which includes both operational and developmental test) objectives. This encompasses such elements as embedded training and testing capabilities, operational training systems, airspace, ranges and off-range lands, scoring and feedback systems, targets, pods/instrumentation/weapon system interface devices, aggressors/contract air support, threat environment generators, networks, synthetic environments, operational training centers, workforce, and cybersecurity. OTI is therefore a sub-set of OTTI and the focus of this AFI.

1.3.5. The TMF provides measurable criteria to apply to development of an integrated, interoperable, and scalable architecture that enables tailored and adjustable threat environments (expanded information on TMF levels is contained in **attachment 6**).

1.4. (Added-ACC) Training System (TS) Acquisition and Sustainment Execution.

1.4.1. **(Added-ACC)** Acquisition and sustainment execution within the Integrated Life Cycle Management (ILCM) enterprise begins with initial requirements identification; continues through development, fielding, operations, and sustainment; and ends with disposal. Acquisition and sustainment execution requires standard processes and business practices to initiate, govern, and/or regulate actions within the ILCM enterprise. See Department of the Air Force Pamphlet (DAFPAM) 63-128, *Integrated Life Cycle Management*

1.4.2. **(Added-ACC)** TS sustainment includes ongoing efforts to maintain concurrency with the prime weapon system to ensure training relevancy and accuracy. Concurrency includes accurate high fidelity functional and operational performance of training device systems and subsystems and all associated training enablers (e.g., Courseware, Instructor Operator Station Instructional Features, Instruction, Net Centric based training features for Live, Virtual, and Constructive (LVC), Distributed Mission Operations (DMO), and Joint Simulation Environment (JSE)).

1.4.3. **(Added-ACC)** For sustainment execution, TSSCs support all units throughout the CAF with TS of the same Major Weapon(s) System (MWS) configuration. See **Attachment 7**, for further details pertaining to TSSC and TSSC Contracting Officer's Representative (COR) duties.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Assistant Secretary of the Air Force for Manpower and Reserve Affairs (SAF/MR): Responsible for military training policy matters.

2.2. Assistant Secretary of the Air Force for Acquisition, Technology, and Logistics (SAF/AQ): Serves as the Air Force Service Acquisition Executive for all USAF programs and oversees all related acquisition programs through the Program Executive Officers. The program executive officers directly influence OTI through the acquisition and sustainment of training systems for USAF warfighters and the USAF's undergraduate aircrew training enterprise.

2.3. Deputy Under Secretary of the Air Force, International Affairs (SAF/IA): Coordinates on policy for international training partners to gain access to USAF OTI and events and provides release authority for classified and unclassified information to foreign training partners.

2.4. Department of the Air Force Chief Information Officer (SAF/CN): Develops, coordinates, and executes USAF information technology policy, strategy, guidance, and oversight for the management of data, information technology systems, and communication networks IAW applicable Office of the Secretary of Defense guidance and DAF 17- & 33-series publications. Chief Information Officer oversight of non-sensitive compartmented information (SCI) networks provides an essential infrastructure capability for distributed operational training by joining distant and disparate training audiences. Approve the Authorizing Official for operational training systems.

2.5. Department of the Air Force Chief Data and Artificial Intelligence Officer (SAF/CND): Develops, coordinates, and executes USAF data and artificial intelligence policy, strategy, guidance, and oversight of USAF data management.

2.6. Department of the Air Force Chief Modeling and Simulation Officer (SAF/CMSO): Provides policy, guidance, and oversight for DAF's modeling and simulation communities.

2.7. Deputy Chief of Staff, Manpower, Personnel, and Services (AF/A1): Develops, coordinates, and executes personnel policy and essential procedural guidance for military training (non-flying) programs.

2.8. Deputy Chief of Staff, Intelligence, Surveillance, Reconnaissance (ISR), and Cyber Effects Operations (AF/A2/6): Develops, coordinates, and executes USAF Intelligence Community information technology policy, strategy, guidance, and oversight for the management of ISR data, SCI ISR networks, and cyber effects operations IAW applicable Office of the Director of National Intelligence guidance and DAF 14-series and 17-series publications.

2.9. Deputy Chief of Staff, Operations (AF/A3) will:

2.9.1. Nominate an Authorizing Official for Operational Training systems to SAF/CN for appointment IAW Department of Defense Instruction (DoDI) 8510.01, *Risk Management Framework (RMF) for DoD Information Technology (IT)*. The Authorizing Official will be a general officer or senior executive service civilian. Authorizing Official responsibilities will be specifically limited to the IT portion of training devices, training aids, models, environment generators, ranges, and systems. The Authorizing Official will designate and ensure the AF/A3

is manned to provide a Security Control Assessor (SCA) and associated SCA Representatives. The SCA is the senior official with the authority and responsibility for the certification of assigned or applicable USAF-governed OTI systems and platform information technology systems. The SCA will work with Air Combat Command Communications Directorate (ACC/A6) for certification for any Operational Training systems that connect to the Air Force Integrated Telecommunications Network. The SCA and SCA Representatives will report to the Authorizing Official.

2.9.2. Act as the training funding advocate for aircrew training systems, missile training systems, cyber training systems, and mission system training systems.

2.9.3. Establish policy and guidance for OTI.

2.10. Chief, Operational Training Infrastructure Division (AF/A3TI) will:

2.10.1. Synchronize operational training system and undergraduate aircrew training system requirements and acquisition strategies with appropriate Secretary of the Air Force (SAF), Air Staff, industry, and program executive officer organizations.

2.10.2. Advocate for operational training systems and undergraduate aircrew training system funding and requirements at SAF, Air Staff, Military Departments, Interagency, and Multi-national processes IAW Air Force Policy Directive (AFPD) 10-6, *Capability Requirements Development*.

2.10.3. Consolidate operational training system and undergraduate aircrew training system funding information from Air Staff functional managers and Program Element Monitors to track USAF-wide operational training system funding status.

2.10.4. Monitor and attend program management reviews, technical interchange meetings, standards working groups, and other training or readiness forums to ensure adherence to established operational training policy and guidance.

2.10.5. Establish policy to ensure sharing of government-owned data, databases, modeling and simulation solutions, and other capabilities to reduce or prevent redundant efforts.

2.10.6. Synchronize MAJCOM efforts to minimize multiple independent USAF networks.

2.10.7. Appoint a training system manager to oversee management of operational training systems and undergraduate aircrew training systems.

2.10.7.1. The training system manager will interface with subject matter experts (SME) across MAJCOMs, total force, joint, and coalition on operational training systems and undergraduate aircrew training systems. Further, the training system manager will analyze MAJCOM reports to provide advocacy to the corporate process.

2.10.7.2. Training system manager will champion the resourcing of concurrency, fidelity, and connectivity of operational training systems and undergraduate aircrew training systems. The training system manager will conduct annual data calls to assess the health of the USAF training system enterprise.

2.10.8. Advocate for OTI requirements, technology, and planning throughout the Joint Capabilities Integration and Development System (JCIDS) IAW Chairman of the Joint Chiefs of Staff Instruction (CJCSI) 5123.01I, *Charter of the Joint Requirements Oversight Council*

(JROC) and Implementation of the Joint Capabilities Integration and Development System, and AFI 10-601, *Operational Capability Requirements Documentation and Validation*.

2.10.9. As co-chair of the OTTI Capability Development Team, advocate for the broader OTTI requirements (see **paragraph 1.3.4.)** and synchronize overlapping developmental and operational test requirements with Capability Development Team co-chair, Air Force Test and Evaluation Programs and Policy (AF/TEP).

2.11. Commander, Air Force Modeling and Simulation Agency (AFAMS) will:

2.11.1. Develop capability-based modeling and simulation (M&S) requirements in support of Training and Readiness.

2.11.2. Oversee all components of the Air, Space, and Cyberspace Constructive Environment, as well as follow-on efforts. This includes providing oversight of user test and integration efforts.

2.11.3. Monitor and attend program management reviews, technical interchange meetings, standards working groups, and other training or readiness forums to champion transformation capabilities.

2.11.4. Synchronize MAJCOM efforts to ensure sharing of government-owned datasets, databases, modeling and simulation solutions, and other capabilities to reduce or prevent redundant efforts.

2.11.5. Provide support to the SAF/CN-designated OTTI Authorizing Official for Risk Management Framework assessments and authorizations.

2.12. Lead Commands Will:

2.12.1. Establish training system management structures to holistically plan, fund, and manage training systems and associated distributed training capabilities (for example, distributed mission operations (DMO)) as part of an OTI framework, to include ensuring training system requirements are included in the Air Force Corporate Structure program objective memorandum process. Lead commands will ensure the major weapon system or prime mission system will not be modified or upgraded unless funding is sufficient to also modify/upgrade all impacted aspects of the operational training system or undergraduate aircrew training system IAW DAFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities*.

2.12.1.1. **(Added-ACC)** Manage TSs using the ILCM outlined in this ACC Supplement and DAFPAM 63-128. For Cross-MAJCOM platforms, adhere to the applicable MAJCOM supplement to this instruction. **(T-2)** TS managers must follow appropriate guidance for acquisition, programming, requirements, testing, and reporting. **(T-2)**

2.12.1.2. **(Added-ACC)** ACC Directorate of Plans, Programs and Requirements (ACC/A5/8/9) platform representative divisions (e.g., Program Element Managers (PEM) at ACC Combat Forces Division (ACC/A5A), ACC F-35 Systems Management Office (ACC/A5/8/9/A5F), ACC Command and Control (C2) Weapons Systems Division (ACC/A5Y)) are responsible for overall prime weapons system requirements, including TS requirements, management, and funding. **(T-2)**

2.12.1.3. **(Added-ACC)** ACC Directorate for Operations (ACC/A3) platform representative divisions (e.g., Functional Area Managers (FAMs) at A3T, ACC Command and Control, Intelligence, Surveillance, and Reconnaissance (C2ISR) Division (ACC/A3C), ACC Persistent Attack and Reconnaissance Division (ACC/A3M)) are responsible for identifying aircrew and mission systems training requirements for TS operations and support of LVC capabilities. **(T-2)**

2.12.1.4. **(Added-ACC)** ACC Test and Training Division (ACC/A5T) and, as appropriate, other ACC divisions are responsible for TS Aircrew Training Device (ATD), and Mission System Training Device (MSTD) requirements (e.g., acquisition, modification/modernization, and sustainment) management. **(T-2)**

2.12.1.5. **(Added-ACC)** ACC Training Support Squadron (TRSS) is responsible for managing assigned Contracted Aircrew Training and Courseware Development contracts. **(T-2)**

2.12.1.6. **(Added-ACC)** ACC will develop Memorandum of Understanding/Memorandum of Agreement with other CAF TS lead commands (LC), using commands, and TS System Program Offices (SPO) (Big Safari, JPO, etc.) to ensure that a consistent application of test processes can be accomplished on all CAF TSs. **(T-2)**

2.12.1.7. **(Added-ACC)** The F-35 JPO leads a US Service/Partner Training System Requirements Team (TSRT) to manage requirements, system configuration, and correct deficiencies.

2.12.2. Manage training requirements by:

2.12.2.1. Establishing aircrew, mission, missile, and cyberspace crew training requirements and document in system training plans (STP) and/or concept documents. The STP is an iterative planning document that defines the justification, design, development, funding, resources, support, modification, operation, and management of a training system. For fielded systems, the STP may be a stand-alone document, or it may be referenced and summarized in the Life Cycle Sustainment Plan or other Human Systems Integration documents. All references to the STP in this document incorporate the possibility that the intended documentation may be part of a Life Cycle Sustainment Plan or Human Systems Integration documents. Training considerations prior to fielding will be addressed in coordination with the prime mission system/weapons system platform or separate training system program manager. Expanded information on the STP is contained in **attachment 3**.

2.12.2.1.1. **(Added-ACC)** A3 divisions are OPR for the STP. **(T-2)**

2.12.2.1.2. **(Added-ACC)** The F-35 is a JPO platform and uses the "F-35 Joint Strike Fighter (JSF) Program Training Infrastructure System Concept of Operations (CONOPS)" in place of an STP. **(T-2)**

2.12.2.2. Defining training system life cycle (development through sustainment) requirements IAW AFPD 63-1/20-1, *Integrated Life Cycle Management*.

2.12.2.3. Establishing training system standards, tasks, and requirements in coordination with Air Force career field managers, functional area managers, MAJCOM functional

managers, training pipeline managers, and Air Education and Training Command (AETC) training managers, when applicable, IAW DAFMAN 36-2689, *Training Program*.

2.12.2.3.1. **(Added-ACC)** Additional SME from ACC TRSS at using command MDS FAM discretion.

2.12.2.3.2. **(Added-ACC)** A3 divisions are responsible for developing, identifying, and approving training tasks documented via a Training Task List (TTL) or similar. A3 divisions are responsible for developing and maintaining TTLs, Formal Training Course syllabi, graduate evaluation data, academic courseware, academic tests, academic instruction, and identifying types of practical and virtual trainers required for formal training courses. **(T-2)**

2.12.2.3.3. **(Added-ACC)** A5 divisions are responsible for maintaining weapons system requirements. **(T-2)** A5 divisions will identify and process documents for TS modifications to support training task requirements. **(T-2)** ***Facility modifications, to accommodate TS changes will be identified to, and coordinated with, the Host Unit for action***.

2.12.2.3.4. **(Added-ACC)** ACC Communications Directorate (ACC/A6) is the OPR for TS cybersecurity requirements. **(T-2)**

2.12.2.3.5. **(Added-ACC)** ACC TRSS is OPR for formal training requirements following the Instructional Systems Development process. **(T-2)**

2.12.2.3.6. **(Added-ACC)** The F-35 JPO develops and maintains common, US Service, and Partner formal training course material.

2.12.2.4. Setting requirements for, and leading training system requirements analysis (TSRA) and STP development and execution, in coordination with Air Force Life Cycle Management Center (AFLCMC), AETC, and using commands, as warranted.

2.12.2.5. Conducting TSRAs, in coordination with using commands, through the prime mission system program office, Training System Program Office (SPO), acquisition program office, or AETC. The TSRA process is designed to derive and allocate training requirements from user identified operational needs. A TSRA integrates the products of the instructional system development process and the systems engineering process to describe the training system to be procured. The TSRA process results in a written record of the compilation and distillation of four complementary elements, including the mission-task analysis, training requirements analysis, media analysis, and the training systems basis analysis. Review standing TSRAs every 5 years, or sooner if changes to the system or training environment dictates, to determine if TSRA modifications are required. The lead command makes final decision on whether a TSRA update or modification will be undertaken. Expanded information on the TSRA is contained in **attachment 2** of this AFI.

2.12.2.5.1. **(Added-ACC)** TSRA process will consider the weapon system, aerodynamic, sensor, weapons performance data, and crew role in the operational environment to determine how much interoperability the TS requires during initial MWS procurement and for follow-on modification and modernization. **(T-2)** TS interoperability will be addressed in all ACC developed TS requirements documents during the definition process. **(T-2)**

2.12.2.6. Maintaining prime mission system requirements, including those validated through the JCIDS process.

2.12.2.7. Reviewing prime mission system, product group, and MAJCOM standards impacting training system requirements, annually.

2.12.2.8. Prioritizing requirements using USAF-approved criteria following Air Force Materiel Command (AFMC) procedures as defined in Air Force Manual (AFMAN) 63-143, *Centralized Asset Management Procedures*.

2.12.2.8.1. **(Added-ACC)** **Attachment 8** describes ACC's TS modification process, including requirements for semiannual Training System and Configuration Working Groups (TSCWG) and Training System Working Groups (TSWG).

2.12.2.9. Establishing MAJCOM policy and requirements that reflect the sharing, validation, and reuse of visual (image generator), computer generated forces (CGF) models, threat databases, weapons modeling, and associated databases, etc. to gain efficiencies and enhance interoperability.

2.12.2.9.1. **(Added-ACC)** For those systems under their control, A5T, in coordination with appropriate A3/A5 divisions, will ensure acquisition and modification documents include appropriate language. **(T-2)** That language will require the maximum reuse and sharing of established databases, models, and other features to increase interoperability and reduce cost. **(T-2)** Additionally, the language will require use of authoritative data sources when available. **(T-2)** TS experts (A3, A5, and Test and Evaluation (T&E) experts) must be included from the start in weapon system development efforts (e.g., meetings, discussions). **(T-2)**

2.12.2.10. Semi-annually report simulator utilization metrics on all contracted weapon system platform training systems. Forward reports to AF/A3TI by 30 April for the prior semi-annual period October through March and by 31 October for the prior semi-annual period April through September. Reference **attachment 4**.

2.12.2.10.1. **(Added-ACC)** A5T or the appropriate A3 division will ensure training device use is reported as defined in **Attachment 4**. **(T-2)**

2.12.2.10.2. **(Added-ACC)** A5T or the appropriate A3 division is responsible for defining TS availability and ensuring acquisition and support documents include operations scheduling and government use requirements. **(T-2)**

2.12.2.10.3. **(Added-ACC)** A5T or the appropriate A3 division will track training device inventory for equipment managed within their area of responsibility. **(T-2)**

2.12.3. Support using commands by:

2.12.3.1. Assisting with development of command requirements and specifications.

2.12.3.1.1. **(Added-ACC)** A3 divisions are responsible for determining unit-level second and third-order effect impacts of institutional training decisions. **(T-2) Note:** This may be done by consulting with unit Operations Groups (OG), Weapon Officers, or via TSWG or TSCWG.

2.12.3.1.2. **(Added-ACC)** Appropriate A3 and A5 divisions will work with the Wing-level supported units to ensure active participation in TS management, acquisition,

modification, T&E, and modernization efforts. **(T-2) Note:** Local units will take part in certain aspects of the training team's effort (i.e., TSWG or TSCWG). **(T-2)**

2.12.3.2. Identifying the best mix of live and synthetic training capabilities (networked or non-networked) to accomplish training requirements.

2.12.4. Support prime mission system and training system program management by:

2.12.4.1. Providing guidance and assistance to program managers in the development, acquisition, and sustainment of training systems and associated distributed training capabilities (for example, DMO).

2.12.4.1.1. **(Added-ACC)** Appropriate A3 and A5 Mission Design Series (MDS) divisions, will provide guidance and assistance to the prime mission system and training system program managers (PM) for TS development, acquisition, and sustainment. This includes defining the scope and support requirements of the TSSC. **(T-2)**

2.12.4.1.2. **(Added-ACC)** The F-35 JPO leads a US Service/Partner TSRT to manage requirements, system configuration, and correct deficiencies.

2.12.4.2. Ensuring the training system is available, network-capable, if applicable, and compliant with required standards prior to fielding the prime mission system.

2.12.4.3. Defining training system requirements for the prime mission system fielding strategy.

2.12.4.3.1. **(Added-ACC)** STP and TSRA documents support the prime mission systems fielding strategy.

2.12.4.4. Notifying the training system program manager of weapon system modifications to ensure training system concurrency.

2.12.4.4.1. **(Added-ACC)** All appropriate ACC divisions will manage TS concurrency requirements, collaborate on prime weapon system modifications, and provide the PM timely notification of concurrency and sustainment modification requirements. **(T-2)** See **Attachment 8**.

2.12.4.5. Coordinating with Air Force Operational Test and Evaluation Center or MAJCOM operational test organizations to guide program managers on test or operational issues for the acquisition and sustainment of training systems.

2.12.4.5.1. **(Added-ACC)** Coordinating with Air Force Operational Test and Evaluation Center (AFOTEC) or MAJCOM operational test organizations to guide PMs on test or operational issues for the acquisition and sustainment of training systems and synthetic environments.

2.12.4.6. Informing program managers of databases, models, and other enterprise capabilities available to increase interoperability and reduce costs.

2.12.4.7. Assessing approved Air Force Forms 1067, *Modification Proposal*, training system impacts and budget estimate with the program manager before the modification proceeds.

2.12.4.7.1. **(Added-ACC)** The F-35 JPO does not rely solely on AF Form 1067 for requirements generation and validation. Modifications to F-35 training systems follow JPO processes.

2.12.4.8. Establishing and maintaining an administrative control process to record and track evaluations of the training system, to include any simulator and synthetic environment certification or simulator validation (see **attachment 5**).

2.12.4.8.1. **(Added-ACC)** A3 divisions are responsible for tracking Simulator Certification (SIMCERT)/Simulator Validation (SIMVAL) operational evaluations and TS accreditation. **(T-2)**

2.12.4.8.2. **(Added-ACC)** The F-35 JPO maintains administrative control processes to record and track TS evaluations. A3T may direct additional evaluations (SIMCERTs/SIMVALs) and reports for ACC use.

2.12.4.9. Providing information and assistance to the program manager for development and modifications to the Systems Engineering Plan, Materiel Fielding Plan, Acquisition Strategy, and other applicable acquisition documents to ensure operational training is appropriately considered in the acquisition process. See AFPD 63-1/20-1, *Integrated Life Cycle Management*, for additional guidance with respect to planning for training in the acquisition process.

2.12.4.9.1. **(Added-ACC)** ACC divisions through A5 counterpart will coordinate with AFLCMC, or the JPO Training Systems and Simulation (TSS) Program Management Office (PMO) for F-35, on TS requirements, funding, facilities, availability, acquisition, modernization, and related sustainment/support. **(T-2)**

2.12.4.9.2. **(Added-ACC)** ACC's simulator OT organization shall be included in planning, accepting, and evaluating capabilities of TS's acquisition testing, evaluating, and Government acceptance capabilities. **(T-2)**

2.12.4.9.3. **(Added-ACC)** Appropriate A3 or A5 divisions (responsible for an MWS Training Device (TD)) will review Operational Training and test methodology submitted by CAF Operational Test Organizations and provide inputs and special interest items to OT agent to aid and focus evaluations. **(T-2)**

2.12.4.9.4. **(Added-ACC)** Appropriate A3 or A5 divisions (responsible for an MWS TD) will consider reported issues for inclusion in updated system requirements. **(T-2)** All other TD test methodology will be IAW 99-series and 53d Wing (53 WG) requirements. **(T-2)**

2.12.4.10. Designating in writing an Information System Security Officer (ISSO) and alternate to ensure cybersecurity compliance.

2.12.5. Manage simulator and synthetic environment certification programs by:

2.12.5.1. Coordinating with appropriate agencies (i.e., using command, AFLCMC and/or product or materiel group) to establish simulator/synthetic environment certification requirements based on training device complexity, desired level of creditable training events, and industry accepted standards, such as Federal Aviation Administration standards, to include training tasks, criteria, and certification interval. Simulator/synthetic environment certifications examine fidelity and usability characteristics to support training

system accreditation and identify capabilities and limitations. Expanded information on simulator/synthetic environment certification is contained in **attachment 5** of this AFI.

2.12.5.1.1. **(Added-ACC)** A3 divisions will process SIMCERT requests through A5T. A5T will manage and coordinate the tasking of the SIMCERT or other evaluation with the 29th Test and Evaluation Squadron (29 TES) for testing IAW ACCI 99-101, *ACC Test and Evaluation*. **(T-2)**

2.12.5.1.2. **(Added-ACC)** A3 divisions are responsible for SIMCERT training tasks and criteria.

2.12.5.2. Ensuring all system types receive simulator/synthetic environment certification at least once every 48 months. Certification procedures and documentation will comply with guidance in USAF 10-series and USAF 11-series publications and MAJCOM guidance. Ensuring simulator/synthetic environment certification procedures confirm that enterprise standard solutions integrated into training systems comply with enterprise standards (for example, motion or visual systems are current with enterprise standard for that model).

2.12.5.2.1. **(Added-ACC)** A3 divisions are responsible for tracking SIMCERT operational evaluations and TS accreditation. **(T-2)**

2.12.5.2.2. **(Added-ACC)** New TSs require a SIMCERT 90 to 120 days after initial fielding. **(T-2) Note:** SIMCERT is event driven and not to be confused with Government Acceptance Testing, which is the verification of contractual requirements.

2.12.5.2.3. **(Added-ACC)** ACC may direct additional SIMCERT, SIMVAL, or other OT evaluation at OT agent recommendation, be accomplished within the 48-month window as required for major modifications, new Operational Flight Plan/Program (OFP), new capabilities, or upon relocation of a device.

2.12.5.3. Allocating and providing funding to accomplish simulator/synthetic environment certification actions.

2.12.5.3.1. **(Added-ACC)** A5 divisions fund and provide guidance for CAF SIMCERTs. A5T will coordinate annual funding requirements with A3T to capture SIMCERT relevant funding requirements applicable to PE 28015F. **(T-2)** A3 divisions will coordinate through A3T to provide SIMCERT relevant funding requirements applicable to PE 28015F to A5T for advocacy purposes. **(T-2)**

2.12.5.3.2. **(Added-ACC)** ACC's SIMCERT program includes assessment/evaluation for CAF-common MWS TSs, if requested. For non-common MWSs, the requesting CAF partner funds the ACC SIMCERT agency's TDY travel. **Note:** Examples of non-common to ACC MWSs are B-1, B-2, B-52, and F-15C.

2.12.5.4. Designating a simulator/synthetic environment certification agent who will:

2.12.5.4.1. Establish deadlines for initial certification of newly delivered devices, upon fielding major modifications, or after relocation of devices.

2.12.5.4.2. Perform a simulator/synthetic environment certification IAW lead command or MAJCOM guidance.

2.12.5.4.3. Document training system simulator/synthetic environment certification status. Lead command guidance will be used to establish further procedures, processes, and requirements. Resources for conducting training system simulator/synthetic environment certification will be provided by the lead command.

2.12.5.4.4. Provide simulator/synthetic environment certification results and the accreditation recommendation to the lead command, or designated accreditation proponent, for review.

2.12.5.4.5. **(Added-ACC)** ACC designates the 53 WG (29 TES) as the SIMCERT operational test agent for CAF simulators. **(T-2)**

2.12.5.4.6. **(Added-ACC)** 29 TES will develop and document SIMCERT Program operational assessment requirements (e.g., test plans, evaluation process, and authority) and reporting format for each MWS TS. **(T-2)**

2.12.5.5. Responding to simulator/synthetic environment certification agent reports within 14 days, through a designated accreditation proponent if desired and appropriate.

2.12.5.5.1. **(Added-ACC)** Appropriate A3 divisions will respond to SIMCERT report as acknowledgment of receipt and task completion and recommend task closure to A5T. **(T-2)**. See **Attachment 5** for additional information regarding accreditation.

2.12.6. Manage simulator validation programs by:

2.12.6.1. Being the validation authority and ensuring CGFs or human-in-the-loop (HITL) for specific training systems are accredited.

2.12.6.1.1. **(Added-ACC)** Being the validation authority and ensuring CGFs or HITL devices for specific training systems or synthetic environments are being accredited.

2.12.6.2. Establishing training system simulator validation requirements and evaluation criteria based on training device complexity, desired TMF level of creditable training events, and industry accepted standards such as Federal Aviation Administration standards, to include training tasks, criteria, and simulator validation interval. Training system simulator validations compare a training device's operating parameters and performance to constructive interactions to ensure highest-fidelity training. Expanded information on simulator validation is contained in **attachment 5** of this AFI. Expanded information on TMF levels is contained in **attachment 6**.

2.12.6.3. Being the validation proponent for training systems procured or sustained by the respective lead command.

2.12.6.4. Allocating and providing funding to accomplish simulator validation actions.

2.12.6.4.1. **(Added-ACC)** A5 platform representative divisions (e.g., Program Element Managers (PEM) at A5A, A5F, A5Y) allocate and provide funding to include manpower resources to accomplish required actions. **(T-2)**

2.12.6.5. Informing using commands about training system changes that will impact accreditation and if a complete or partial simulator validation is required.

2.12.6.6. Using the training system simulator validation control process to record training system simulator validation results related to CGF configurations.

2.12.6.7. Reviewing the results of simulator validations and ensure CGF accreditation for the specific training system.

2.12.6.8. Designating a simulator validation agent. The agent will perform validations as directed by the lead command and provide results to the lead command for accreditation approval.

2.12.6.8.1. **(Added-ACC)** ACC designates the 53 WG (29 TES) as the SIMVAL operational test agent for CAF simulators/synthetic environments. **(T-2)**

2.12.6.9. Ensuring all training system types (flight, mission, part-task, etc.) receive simulator validation at least once every 48 months. Validation procedures and documentation will comply with guidance in USAF 16-series and USAF 10-series publications.

2.12.6.9.1. **(Added-ACC)** A3 divisions are responsible for tracking SIMVAL operational evaluations. **(T-2)**

2.12.6.9.2. **(Added-ACC)** ACC may direct additional SIMCERT, SIMVAL, or other OT evaluation at OT agent recommendation, be accomplished within the 48-month window as required for major modifications, new OFP, new capabilities, or upon relocation of a device.

2.12.7. Manage training system use, distribution, and disposition.

2.12.7.1. **(Added-ACC)** A5 divisions plan, budget, program, and manage concurrency, obsolescence, updates, modifications, distribution, and disposition for TSs. **(T-2)**

2.12.8. Provide resources required to obtain and maintain cybersecurity certification and Authority to Operate, and Authority to Connect.

2.12.8.1. **(Added-ACC)** A5 divisions plan, budget, program and manage resources to maintain cybersecurity certification, Authority to Operate, and Authority to Connect. **(T-2)**

2.12.9. Designate SMEs for the training system.

2.12.10. Provide SME support for contract performance evaluations and government acceptance testing.

2.12.10.1. **(Added-ACC)** A3 divisions will coordinate SME support with OT agent for all CAF performance evaluations and acceptance testing activities. **(T-2)**

2.12.11. In coordination with the program manager and using command, provide representation at source selections, technical reviews and audits, program management reviews, technical interchange meetings, contract award conferences, training system working groups, testing and logistics interchange meetings, and facility design and construction meetings.

2.12.11.1. **(Added-ACC)** A3/A5 divisions and ACC units will participate at source selections, technical reviews and audits, program management reviews, technical interchange meetings (TIM), contract award conferences, TS working groups, testing and logistics interchange meetings, and facility design and construction meetings. **(T-2)**

Appropriate A3/A5 divisions will coordinate Operational Test Agency-requested SME support for all evaluations and testing. **(T-2)**

2.12.12. Chair the Training Planning Team (TPT).

2.12.12.1. **(Added-ACC)** A3 divisions chair the Training Planning Team (TPT). **(T-2)**

2.12.13. Define the scope and support requirements of the Training System Support Center (TSSC), as required.

2.12.13.1. **(Added-ACC)** A5 divisions are responsible for defining the scope and support requirements of the TSSC to its MDS. **(T-2)** The JPO has established a system of US Service and Partner TSSCs to support F-35 training systems.

2.12.14. Develop, maintain, and review the STP for all lead-command major weapon systems.

2.12.14.1. **(Added-ACC)** A3 divisions will develop, maintain, review, and approve STPs for prime mission systems. **(T-2)** The LC can delegate STP review/update duties.

2.12.14.2. **(Added-ACC)** Review and update the STP annually throughout the life cycle of the prime mission system.

2.12.14.3. **(Added-ACC)** Review changes affecting training that have occurred in mission tasking, threat capabilities, tactics, training audience experience level and background, TS availability, component capabilities, funding priorities, basing, new training technologies, and training deficiencies identified in mishap reports.

2.12.15. Ensure the training system is included in the major weapon system modification budgeting profile, maintains concurrency, cybersecurity accreditation, addresses diminishing manufacturing sources and material shortages, and complies with networked training and synthetic environment integration standards.

2.12.16. Ensure future aircraft operational training systems used for synthetic high-end advanced tactics and training integrate with the Joint Simulation Environment (JSE).

2.12.16.1. **(Added-ACC)** ACC is the lead command for the development of JSE. A3T represents the using community for the gathering and initial prioritization of requirements for A5T management. A5T is responsible for the development and sustainment of common system attributes within the environment. **(T-2)** Appropriate A3 and A5 MDS divisions are responsible for development and sustainment of TSs to integrate with JSE as required. **(T-2)**

2.12.17. Accomplish biennial program reviews of each legacy training system used for high-end tactics and training to determine the appropriate degree of JSE integration. At a minimum, future modifications to legacy high-end training systems should consider JSE compatibility.

2.12.18. Provide guidance to fund and provide Contracting Officer's Representative (COR) positions, on-site and on-call, for their assigned training enterprises. The COR performs oversight for the contracting officer.

2.12.18.1. **(Added-ACC)** The F-35 JPO provides CORs for JPO contracted training systems and operations.

2.12.18.2. **(Added-ACC)** Appropriate A5 division will provide guidance to fund and provide COR positions, on-site and on-call, for their assigned training enterprises. **(T-2)**

The COR performs oversight for the contracting officer. **(T-2)** Manpower billets generated to support a training system will remain assigned to that device and will not be used for other purposes in the unit. **(T-2)**

2.12.18.3. **(Added-ACC)** ACC units with on-site TS contractor logistics support (CLS) will provide a COR to perform contract oversight. **(T-2)** The Unit COR is empowered by the contracting officer and is the sole on-site interface between government personnel and the contractor. For emerging MWSs, the A5 divisions are OPR for programming TS COR positions. **(T-2)** The Unit COR should be assigned to the unit organization (e.g., Operations Support Squadron). See **Attachment 7**, and **Attachment 8**, for further details.

2.12.18.4. **(Added-ACC)** ACC host unit OG/CC is the Functional Commander (FC)/Functional Director (FD) for sites possessing TSs. FC/FD duties may be delegated to the unit.

2.12.18.5. **(Added-ACC)** In addition to the duties authorized by the contracting officer's Letter of Delegation (LOD), the FC/FD may require CORs to perform additional duties. However, additional duties will not interfere with COR primary contract oversight duties and responsibilities. **(T-2)**

2.12.18.6. **(Added-ACC)** ACC programs with a TSSC will have a TSSC COR to perform contract oversight. **(T-2)** The TSSC COR provides surveillance over the quality of a contractor's software development, modification, and support at the TSSC to ensure the contractor has performed IAW contract requirements. **(T-2)** The TSSC COR is also the central point of contact for TSSC contract administration, logistic, and security support as delegated. **(T-2)** A5 divisions are OPR for programming TSSC COR positions and will program additional billets for all TS programs. **(T-2)** The TSSC COR duties should be assigned to the 29 TES at locations where a TSSC is located on a military installation co-located with a 29 TES Detachment or Operating Location (OL). For locations where a TSSC is not located on a military installation, remote TSSC CORs may be used to achieve required government oversight with contracting officer approval. **(T-2)** See **Attachment 7**. **Note:** For ACC units with a TRSS, COR written guidance related to contract management is contained in ACCI 11-251 or in each individual Performance Work Statement.

2.12.19. Unit manning document billets generated to support a training system will remain assigned to that system and will not be re-assigned to perform other purposes in the unit. Personnel filling that billet may perform additional duties on a non-interference basis.

2.12.20. Report training system data on all assigned weapon system platforms to the AF/A3TI training system manager no later than 31 January (annually). Specific requirements will be detailed in the AF/A3TI tasking message (see **paragraph 3.3.1**.).

2.12.21. Take appropriate program objective memorandum actions to replace existing orphan training systems before those training systems can no longer meet training needs due to obsolescence or serviceability**.**

2.12.22. Provide concepts of operation, force development concepts, operational roadmaps, and applicable operations training requirements.

2.12.23. Provide updates on training capabilities and infrastructure development which impact training system planning.

2.12.24. Implement a training device fleet flying hour program. Provide "Available Hours Contracted for Use" from Training System Metrics (Item C from **Figure A4.1**), which are reported semi-annually to AF/A3TI, to AFLCMC on the same schedule for Operational Cost per Training System Flying Hour calculations.

2.12.25. Ensure safety requirements for training systems comply with DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*.

2.13. Using Commands will:

2.13.1. Identify training and operational training system requirements to the lead command for advocacy, programming, and funding.

2.13.2. Designate in writing an ISSO and alternate to ensure cybersecurity compliance.

2.13.3. Submit reporting information to the lead command as directed in this AFI.

2.13.4. Identify technology gaps and operational requirements to the lead command to facilitate sustainment support and identify trainer sustainment requirements in conjunction with all mission design series and training system requirements.

2.13.5. Assume lead command responsibilities for simulator/synthetic environment certification and simulator validation when the operational training system is unique to a respective using command. Simulator/synthetic environment certification and simulator validation procedures will comply with guidance in AFI 16-1001, *Verification, Validation and Accreditation (VV&A)*.

2.13.5.1. **(Added-ACC)** Establish TS accreditation process for those TSs that are not part of ACC's inventory or have additional mission requirements different from ACC's. **(T-2)**

2.13.5.2. **(Added-ACC)** Provide SMEs to support TS acquisition, test and evaluation, and modification activities (e.g., requirements definition, design reviews, performance assessments and evaluations, formal testing and acceptance, simulator certification, simulator validation). **(T-2)**

2.13.6. Fund repairs on all command-unique training devices not associated with prime mission systems, including obsolescent systems where training system parts are no longer procurable or available through original equipment manufacturer.

2.14. MAJCOMs, National Guard Bureau, Direct Reporting Units, and Field Operating Agencies will: Assume both using command and lead command responsibilities for training systems developed or acquired by those commands to meet unique training needs.

2.14.1. **Air Education and Training Command (AETC) will:**

2.14.1.1. Act as the USAF's Executive Agent for formal, undergraduate training programs technology and development, for which AETC is the lead command.

2.14.1.2. Advocate for distributed training/remote instruction capabilities in formal education and training under its command.

2.14.1.2.1. Ensure training systems requirements are addressed from the start of the program by membership on the TPTs with other system stakeholders as early in the system concept as possible.

2.14.1.2.2. Ensure AETC TPT participation directly supporting SME participation for the development of the STP and TSRA.

2.14.1.2.3. Establish a command position for training system requirements and resources required to meet AETC's training requirements.

2.14.1.2.4. Establish AETC TPT support teams to develop, support, and maintain each training system.

2.14.1.2.5. Monitor new acquisitions, modifications, or changes to the weapon system managed by the acquisition SPO that affect training systems and equipment. These SPO activities include fielded training systems that require the development of requirements documents IAW AFI 10-601.

2.14.1.3. Provide advice and expertise to the lead command, program manager, and TPTs.

2.14.1.4. Perform TSRAs, when requested by the lead command, in coordination with the TPT.

2.14.1.5. Semi-annually report simulator utilization metrics on all contracted undergraduate aircrew training systems. Forward reports to AF/A3TI by 30 April for the prior semi-annual period October through March and by 31 October for the prior semi-annual period April through September. Reference **Attachment 4**.

2.14.1.6. Accomplish simulator/synthetic environment certification of undergraduate training aircraft simulators IAW **paragraph 2.12.5** and **attachment 5**.

2.14.1.7. Report training system data on all assigned undergraduate aircrew platforms to the AF/A3TI training system manager no later than 31 January (annually). Specific requirements will be detailed in the AF/A3TI tasking message (see **paragraph 3.3.1**.).

2.14.2. **Air Force Materiel Command (AFMC) will:**

2.14.2.1. Coordinate lead command and prime mission system program offices' training system requirements and solutions. Provide support to SPOs responsible for training system acquisition and sustainment through the parent prime mission systems program office(s) and/or AFLCMC.

2.14.2.2. With AFLCMC and Air Force Research Laboratory (AFRL) Warfighter Readiness Research Division, provide technical support to AF/A3TI.

2.14.2.3. Manage sustainment fund programming and allocation through AFMC centralized asset management.

2.14.2.4. Ensure prime mission system modification development in tandem with the training system to ensure concurrency.

2.14.3. **Air Force Life Cycle Management Center (AFLCMC) will:**

2.14.3.1. Perform training system contract management, support (including TSSC establishment), relocation, and disposition activities.

2.14.3.1.1. **(Added-ACC)** The JPO has the responsibility to perform F-35 training system contract management, support, relocation, and disposition activities.

2.14.3.2. Host an All-Command simulator/training device summit to review training system status and discuss lessons learned, potential program synergies, training opportunities, advocacy issues, and future technology needs. Consider holding the summit in conjunction with a training system industry day.

2.14.3.3. Through the simulator SPO, acquisition program office, or AETC, and in coordination with the lead command, conduct TSRAs.

2.14.3.4. Assume Planning, Programming, Budgeting, and Execution System responsibility for active contractor logistics support, and operations and maintenance.

2.14.3.4.1. **(Added-ACC)** The JPO has the responsibility for F-35 Program & Budget Enterprise System execution to support CLS, operations, and maintenance.

2.14.3.5. Provide programmed contractor logistics support to all fielded operational training systems, including programs of record and interim devices.

2.14.3.5.1. **(Added-ACC)** The JPO has the responsibility for F-35 programmed CLS to all fielded TSs.

2.14.3.6. Lead training system research and development and develop or assist with use case demonstrations and training program and exercise integration.

2.14.3.7. Work with Lead MAJCOMs to ensure training systems comply with architecture standards and maintain common interfaces for integration into high-end synthetic environments to support large force events.

2.14.3.8. Support JCIDS documentation and supporting analyses, technical documentation and validation regarding training efforts, and training system and support service integration and procurement.

2.14.3.9. Ensure the ISSO assists the Program Management Office-appointed Information System Security Manager to complete Risk Management Framework Assessment and Authorization packages and serve as the focal point for cybersecurity compliance involving networked (for example, DMO) and non-networked activities.

2.14.3.9.1. **(Added-ACC)** The JPO has the responsibility to complete Risk Management Framework Assessments and Authorizations packages.

2.14.3.10. Provide resources required to obtain and maintain cybersecurity certification and Authority to Operate.

2.14.3.10.1. **(Added-ACC)** The JPO has the responsibility to provide resources to obtain and maintain cybersecurity certification and Authority to Operate.

2.14.3.11. Capture Operations & Support costs from current contracts to the extent practical. Establish and maintain a methodology for Materiel Leaders and Lead Commands to jointly report annual Operational Cost per Training System Flying Hours by training device fleet.

2.15. Training Planning Team (TPT) will:

2.15.1. Develop the STP after the TSRA is completed.

2.15.2. Provide guidance on master task list content, as warranted.

2.15.3. Use the STP to ensure training considerations are adequately addressed in the prime mission system acquisition and modification processes. The STP is an iterative planning document that defines the justification, design, development, funding, resources, support, modification, operation, and management of an operational training system. The STP also provides instructional considerations and usage to maximize use as an effective educational tool. Expanded information on the STP is contained in **Attachment 3** of this AFI.

2.16. Training System Support Center (TSSC). The TSSC will provide configuration management functions, engineering development, feasibility studies, life cycle hardware, software, and courseware support for the operational training system.

2.17. Host Unit. The host unit (the unit with primary responsibility for the systems) will:

2.17.1. Provide guidance for the unit's training system and assign personnel for COR duty, on-site and on-call, for their assigned training system. **(T-2)**

2.17.2. Fund travel for training system training and COR certification training. **(T-2)**

2.17.3. Provide personnel to support training system testing. **(T-2)**

2.17.4. Maintain training system simulator utilization records and report utilization data to lead command monthly (supports lead command requirement to report simulator utilization data IAW **paragraph 2.12.2.10** and **attachment 4**). **(T-2)**

2.17.5. Debrief maintenance technicians after each training period and provide documentation of all discrepancies in the method prescribed by the system sustainment plan or contract. **(T-2)** Missile procedures training device discrepancy forms may be developed locally.

2.17.6. Ensure the ISSO assists the Program Management Office-appointed Information System Security Manager to complete Risk Management Framework Assessment and Authorization packages and serve as the focal point for cybersecurity compliance involving networked (for example, DMO) and non-networked activities. **(T-0)**

2.17.7. Ensure local communications support for distributed training capabilities. **(T-2)**

2.17.8. Secure training system facilities to the extent necessary to allow for required classified training. **(T-2)**

2.17.9. Ensure facilities are accredited and maintain security accreditation for the simulator/training device and training operations hosted within the facility.

2.18. (Added-ACC) United States Air Force Warfare Center (USAFWC) will:

2.18.1. **(Added-ACC)** In coordination with Headquarters Air Force, Operational Training Infrastructure Division (AF/A3TI) and ACC Flight Operations and Training Branch (ACC/A3TO), the United States Air Force Warfare Center (USAFWC), as Office of Coordinating Responsibility, will be responsible for shaping and influencing the Operational Training and Test Infrastructure (OTTI). **(T-2)** This will include simulators, the battlespace environment, threat and weapon modeling fidelity, visual databases, DMO/simulator

networking and integration, and the ability to meet CAF Operational Training requirements, including the development and assessment of all CAF Maintenance and Support operations. **(T-2)**

2.18.2. **(Added-ACC)** In conjunction with the OTTI Core Function Team, the USAFWC will host an annual review of OTTI including CAF synthetic Operational Training Infrastructure and provide requirements and recommendations to A3T and A5T for incorporation into the STPs and Training Planning Team. **(T-2)**

2.19. (Added-ACC) Local Units (the units that use the TS for training) will:

2.19.1. **(Added-ACC)** Ensure supporting units will meet minimum qualifications as identified by the operational testing organization to accomplish either simulator certification, validation, or other testing (e.g., 1 SEFE/4-ship FL, Instructor). **(T-2)**

2.19.2. **(Added-ACC)** Tailor ATD profiles to meet unit training requirements. Training requirements involving an update to the configuration baseline must be submitted to the TSCWG to formalize and prioritize the change for TSSC or other action. **(T-2)**

2.19.3. **(Added-ACC)** Assign ATD scheduling responsibilities to their scheduling office as determined by the OG/CC. **(T-2)**

2.19.4. **(Added-ACC)** Assign System Administrators (SYS AD) and POCs appointed by the OG/CC, when required. **(T-2)** POCs and/or SYS ADs arrange for maintenance support with a designated TS site COR, or as directed by the PM, for unit simulators supported by overarching on-call CLS. **(T-2)**

2.19.5. **(Added-ACC)** Select the MSTD COR by the OG/CC, when required. **(T-2)**

Chapter 3

PLANNING AND REPORTING REQUIREMENTS

3.1. Advocacy. AF/A3TI supports readiness by ensuring OTI addresses training requirements balanced with affordability. Training system status information from the field is used to advocate in support of OTI capabilities and readiness.

3.2. Planning. Lead commands accomplish training system planning to ensure training requirements are met and adjusted over time for changes that occur. The process evaluates training system requirements against current and projected training system capabilities to identify shortfalls, and then to guide sustainable development to mitigate these shortfalls. This process results in lead command plans for training systems that are formulated and updated on a recurring basis.

3.2.1. Lead commands will ensure that all appropriate forums consider all training systems. **(T-1)**

3.2.2. As a minimum, operational training systems will be included in all training review activities. **(T-1)** Undergraduate aircrew training systems should be included in all undergraduate training review activities. Training review boards and Weapons and Tactics Conferences are leveraged to maximize the quality of unit training and convene (usually annually) to update training events, frequency, and standards.

3.2.3. When training roadmaps are updated, lead commands will ensure the updated documents are delivered to AF/A3TI. **(T-1)** Roadmaps should include training systems location, prime mission, training system modifications over the next 5 years, and known shortfalls or funding disconnects.

3.3. Reporting.

3.3.1. AF/A3TI will conduct one data call per year to gather detailed information on the training system enterprise; specific requirements will be detailed in the tasking message. AF/A3 will use the data collected to assess the synthetic training enterprise against established OTI metrics based on the TMF (expanded information on TMF levels is contained in **Attachment 6**).

3.3.2. MAJCOMs will semi-annually report simulator utilization metrics as prescribed in **paragraph 2.12.2.10.**, **paragraph 2.14.1.5**, and in **attachment 4**. **(T-1)**

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Deputy Chief of Staff, Operations

(ACC)

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Deputy Director of Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

CJCSI 5123.01I, *Charter of the Joint Requirements Oversight Council and Implementation of the Joint Capabilities Integration and Development System,* 30 October 2021

DoDI 5000.02, *Operation of the Adaptive Acquisition Framework,* 23 January 2020, Change 1, 8 June 2022

DoDI 8510.01, *Risk Management Framework (RMF) for DoD Information Technology (IT)*, 19 July 2022

DAFPD 10-2, *Readiness*, 20 May 2021

DAFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities*, 25 May 2021

(Added-ACC) DAFI 20-101/63-101, *Integrated Life Cycle Management*, 16 February 2021

DAFMAN 36-2689, *Training Program*, 31 March 2023

DAFMAN 90-161, *Publications Processes and Procedures,* 18 October 2023

DAFMAN 91-203, *Air Force Occupational Safety Fire and Health Standards*, 25 March 2022

(Added-ACC) DAFPAM 63-128, *Integrated Life Cycle Management*, 3 February 2021

AFPD 10-6, *Capability Requirements Development*, 6 November 2013

AFPD 63-1/20-1, *Integrated Life Cycle Management*, 7 August 2018

AFI 10-601, *Operational Capability Requirements Documentation and Validation,* 27 April 2021

AFI 16-1001, *Verification, Validation and Accreditation (VV&A)*, 29 April 2020

AFI 33-322, *Records Management and Information Governance Program*, 23 Mar 2020

AFMAN 63-143, *Centralized Asset Management Procedures*, 18 December 2020

(Added-ACC) ACCI 99-101, *ACC Test and Evaluation*, 24 August 2020

Adopted Forms

AF Form 1067, *Modification Proposal*

DAF Form 847, *Recommendation for Change of Publication*

(Added-ACC) AFTO Form 244, *Industrial/Support Equipment Record*

(Added-ACC) AFTO Form 781A, *Maintenance Discrepancy and Work Document*

Abbreviations and Acronyms

(Added-ACC) ACC—Air Combat Command

AETC—Air Education and Training Command

(Added-ACC) AFB—Air Force Base

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

(Added-ACC) AFOTEC—Air Force Operational Test and Evaluation Center

AFPD—Air Force Policy Directive

AFRC—Air Force Reserve Command

AFRL—Air Force Research Laboratory

ANG—Air National Guard

AOC—Air Operations Center

(Added-ACC) ATD—Aircrew Training Device

(Added-ACC) CAF—Combat Air Forces

(Added-ACC) CCB—configuration control board

CGF—Computer Generated Forces

(Added-ACC) CLS—Contractor Logistics Support

(Added-ACC) CONOPS—Concept of Operations

COR—Contracting Officer's Representative

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

(Added-ACC) DAFPAM—Department of the Air Force Pamphlet

DAFPD—Department of the Air Force Policy Directive

DMO—Distributed Mission Operations

(Added-ACC) DoD—Department of Defense

DoDI—Department of Defense Instruction

(Added-ACC) ECP—Engineering Change Proposal

ESOH—Environment, Safety, and Occupational Health

(Added-ACC) FAM—Functional Area Manager

(Added-ACC) FC—Functional Commander

(Added-ACC) FD—Functional Director

HITL—Human-In-The-Loop

(Added-ACC) HQ—Headquarters

IAW—In Accordance With

(Added-ACC) ILCM—Integrated Life Cycle Management

ISR—Intelligence, Surveillance and Reconnaissance

ISSO—Information System Security Officer

IT—Information Technology

JCIDS—Joint Capabilities Integration and Development System

JPO—Joint Program Office

JSE—Joint Simulation Environment

(Added-ACC) JSF—Joint Strike Fighter

JTAC—Joint Terminal Attack Controller

(Added-ACC) LC—Lead Command

(Added-ACC) LOD—Letter of Designation

(Added-ACC) LVC—Live, Virtual, and Constructive (refers to Capabilities)

M&S—Modeling and Simulation

MAJCOM—Major Command

MDS—Mission Design Series

(Added-ACC) MSTD—Mission System Training Device

(Added-ACC) MTC—Mission Training Center

(Added-ACC) MWS—Major Weapon(s) System

(Added-ACC) OFP—Operational Flight Plan/Program

(Added-ACC) OG/CC—Operations Group Commander

(Added-ACC) OG—Operations Group

(Added-ACC) OL—Operating Location

(Added-ACC) OPR—Office of Primary Responsibility

(Added-ACC) OT—Operational Training

OTI—Operational Training Infrastructure

OTTI—Operational Training and Test Infrastructure

(Added-ACC) PEM—Program Element Manager

(Added-ACC) PE—Program Element

PESHE—Programmatic Environment, Safety, and Occupational Health Evaluation

(Added-ACC) PMO—Program Management Office

(Added-ACC) PM—Program Manager

(Added-ACC) QASP—Quality Assurance Surveillance Plan

(Added-ACC) RAP—Ready Aircrew Program

(Added-ACC) ROC—ACC Rehearsal of Concept

(Added-ACC) RTM—Ready Aircrew Program Tasking Memorandum

SAF—Secretary of the Air Force

(Added-ACC) SCAP—System Capability Analysis Plan

(Added-ACC) SCAR—System Capability Analysis Report

SCA—Security Control Assessor

SCI—Sensitive Compartmented Information

(Added-ACC) SEFE—Standardization Evaluation Flight Examiner

(Added-ACC) SIMCERT—Simulator Certification

(Added-ACC) SIMSPO—Simulator Program Office

(Added-ACC) SIMVAL—Simulator Validation

SME—Subject Matter Expert

SPO—System Program Office

STP—System Training Plan

(Added-ACC) SYS AD—System Administration

(Added-ACC) T&E—Test and Evaluation

(Added-ACC) TD—Training Device (e.g., MTC, FMT, etc.)

(Added-ACC) TES—Test and Evaluation Squadron

(Added-ACC) TIM—Technical Interchange Meeting

TMF—Threat Matrix Framework

TPT—Training Planning Team

(Added-ACC) TRSS—Training Support Squadron

(Added-ACC) TSCWG—Training System Configuration Working Group

TSRA—Training System Requirements Analysis

(Added-ACC) TSRT—Training System Requirements Team

TSSC—Training System Support Center

(Added-ACC) TSS—Training Systems and Simulation

(Added-ACC) TS—Training Systems

(Added-ACC) TTL—Training Task List

USAF—United States Air Force

(Added-ACC) USAFWC—United States Air Force Warfare Center

USSF—United States Space Force

(Added-ACC) US—United States

Office Symbols

(Added-ACC) ACC/A3—ACC Directorate for Operations

(Added-ACC) ACC/A3C—ACC C2ISR Division

(Added-ACC) ACC/A3M—ACC Persistent Attack and Reconnaissance Division

(Added-ACC) ACC/A3T—ACC Flight Operations Division

(Added-ACC) ACC/A3TO—ACC Flight Operations and Training Branch

(Added-ACC) ACC/A5/8/9/A5F—ACC F-35 Systems Management Office Division

(Added-ACC) ACC/A5—ACC Combat Forces Division

(Added-ACC) ACC/A5T—ACC Test and Training Division

(Added-ACC) ACC/A5TT—ACC Test and Evaluation Branch

(Added-ACC) ACC/A5Y—ACC Command and Control (C2) Weapons Systems Division

ACC/A6—Air Combat Command Communications Directorate

AF/A1—Deputy Chief of Staff, Manpower, Personnel, and Services

AF/A2/6—Deputy Chief of Staff, Intelligence, Surveillance, Reconnaissance, and Cyber Effects Operations

AF/A3—Deputy Chief of Staff, Operations

AF/A3TI—Chief, Operational Training Infrastructure Division

AF/A3T—Training and Readiness Directorate

AF/IO—F-35 Integration Office

AF/TEP—Air Force Test and Evaluation Programs and Policy

AFAMS—Air Force Agency for Modeling and Simulation

(Added-ACC) SAF/AA—Administrative Assistant to the Secretary of the Air Force

SAF/AQ—Assistant Secretary of the Air Force for Acquisition, Technology, and Logistics

SAF/CMSO—Department of the Air Force Chief Modeling and Simulation Officer

SAF/CND—Department of the Air Force Chief Data and Artificial Intelligence Officer

SAF/CN—Department of the Air Force Chief Information Officer

SAF/IA—Deputy Under Secretary of the Air Force, International Affairs

SAF/MR—Assistant Secretary of the Air Force for Manpower and Reserve Affairs

Terms

Accreditation—Official determination that the operational training system is acceptable for the specific training tasks or events to be accomplished in the device.

Air Force career field managers—Individuals appointed by their respective Headquarters Air Force Deputy Chief of Staff or director to ensure development, implementation, and maintenance of career field education and training plans for their assigned Air Force specialties. Air Force career field managers communicate directly with MAJCOM functional manager, Air Reserve Component, and AETC training pipeline manager to disseminate Air Force and career field policies and program requirements.

Aircrew training device—A training device used to prepare aircrew members for the actual performance of aircrew duties. Aircrew training devices can include training devices such as: cockpit familiarization trainers, cockpit procedures trainers, operational flight trainers, part task trainers, fuselage trainer, boom-operator weapon system trainer, and weapon system trainers, just to name a few.

Centralized asset management—AFMC's Program Office process to develop and manage programs using the following four main pillars: Centralized sustainment funding, logistics requirements determination, performance-based logistics, and integrated wholesale supply and depot maintenance operations.

Computer generated forces—A generic term used to refer to computer representations of forces in models and simulations that attempts to model human behavior sufficiently so that the forces will take some actions automatically (without requiring human-in-the-loop interaction). Types of CGF include automated forces - CGFs that require little or no human interaction; semi-automated forces - CGFs in which the individual platform simulation is operated by computer simulation of the platform crew and command hierarchy. (IEEE 1278.3-1996)

Concurrency—The condition where the configuration and operation of the training system matches the configuration and functionality of the reference prime mission system(s), to the extent necessary to provide required training. For training devices, this condition includes the operational [flight] program, mission software, weapons, hardware, and third-party systems that sufficiently and accurately reflects the current configuration of the prime mission system(s) functionality.

Connectivity—The capability of connecting to secure networks to conduct local and long-haul training with other players (shooters, command and control, Intelligence, Surveillance, and Reconnaissance capabilities, and so forth), based at other locations including network training centers (for example, distributed training centers).

Contracting officer's representative—An individual who is designated and authorized in writing by the Contracting Officer to perform specific technical or administrative functions on contracts or orders. This individual may perform the duties of a contractor logistics support Contracting Officer's Representative or a TSSC Contracting Officer's Representative.

Contractor logistics support—A support concept where a contractor is used to provide logistics support for a system, subsystem, modification, or equipment. Contractor logistics support covers activities equivalent to depot maintenance and, as negotiated with the using command, necessary organizational and intermediate level maintenance, software support, and other operation and maintenance tasks.

Courseware—All instructional material including technical data, textual materials, audio tapes, slides, movies, video tapes, video discs, and other audiovisual materials.

Distributed mission operations—The networking of warfighter training that utilizes the integration of virtual and constructive entities, systems, and environments via secure wide-area network to acquire and sustain mission essential competencies required for operational readiness. Distributed mission operations expand a unit's training capabilities and resources to facilitate inter-team training among geographically separated and composite force teams to execute major weapon system training tables, ready aircrew program activities, mission rehearsals, tactics, techniques, and procedures training, and large force exercises.

Distributed Training Center—A center for operational management of distributed training activity and capabilities and a portal for connectivity with and/or management of training activity with other entities.

Fidelity—The degree to which the synthetic environment (for example, visuals, databases, threats, CGFs, weapons performance, aero modeling, and so forth) correctly represents the real-world environment to the operator to accomplish mission training. Operational training models need to support a high-fidelity simulation capability. Access to accurate and representative models and simulation is crucial, and all entities should be available and simulated to the appropriate level of detail. The quality of models directly determines the quality of simulation products.

Formal training—Training in an officially designated course conducted or administered IAW appropriate course outline and training objectives.

Full operational capability—The full attainment of the capability to effectively employ a weapon system, item of equipment, or system of approved specific characteristics, which is manned and operated by a trained, equipped, and supported military unit or force. Full operational capability is not necessarily a date; it defines the criteria necessary to declare full operational capability.

High-end advanced tactics and training—The highest level of tactics development and training that can be accomplished by a platform in a controlled venue (live, synthetic, and/or blended) to meet its expected TMF level mission environment. Normally, this is also the venue where the highest level of instructor training or operational readiness validation is accomplished. This is also where Tactics, Techniques, and Procedures (TTP) development or validation occurs for specific platforms based upon their expected threat environments or new employment concepts.

Human in the loop simulation—A model type that requires human interaction during runtime. It employs one or more human operators in direct control of the simulation/simulator or in some key support function. Human-centered M&S is distinguished from science and process-based simulations where human intervention and modeling does not occur. (DoD M&S Book of Knowledge)

Human systems integration—The process of effective integration of manpower, personnel, training, human factors, safety and occupational health, personal survivability, and habitability considerations into the acquisition of prime mission systems to improve total system performance and reduce costs by focusing attention on the capabilities and limitations of humans.

Initial operational capability—That first attainment of the capability to employ a weapon effectively, item of equipment, or system of approved specific characteristics with the appropriate number, type, and mix of trained and equipped personnel necessary to operate, maintain, and

support the system. It is normally defined in the capability development document. **Note:** initial operational capability is event-driven and not tied to a specific future date.

Instructional system development—A deliberate and orderly process for planning and developing instructional programs that make sure personnel are taught the knowledge, skills, and attitudes essential for successful job performance. It depends on a description and analysis of the tasks necessary for performing the job, objectives, and tests clearly stated before instruction begins, evaluation procedures to determine whether the objectives have been reached, and methods for revising the process based on empirical data.

Interoperability—The ability of systems, units, or forces to provide data, information, materiel, and services to, and accept the same from, other systems, units, or forces, and to use the data, information, materiel, and services exchanged to enable them to operate effectively together. Information Technology (IT) interoperability includes both the technical exchange of information and the end-to-end operational effectiveness of that exchange of information as required for mission accomplishment. Interoperability is more than just information exchange. It includes systems, processes, procedures, organizations, and missions over the life cycle and must be balanced with cybersecurity.

Lead command—The MAJCOM that is the primary operator of a system, subsystem, or item of equipment. This designation generally applies to those operational commands or organizations designated by Headquarters Air Force to conduct or participate in operations or operational testing (see DAFPD 10-9). The USAF assigns responsibility for overall management of each system, subsystem, or item of equipment to a lead command. The lead command contributes to the process of developing and maintaining a force structure with a balance of complementary capabilities, and it establishes a basis for rational allocation of scarce resources among competing requirements. The lead command is responsible for advocating, programming, and allocating funding for those systems assigned to it.

(Added-ACC) Major Weapons System (MWS)—A weapon system acquired through a major defense acquisition program, (e.g., F-35, F-15, C-17, B-1).

Master task list—Documentation of total tasks that may be performed on a weapon system. These lists may include the entire spectrum of tasks in each functional area (operations, maintenance, and support). The master task list provides a task baseline enabling further analysis and the formulation of formal training decisions.

Milestone A—An investment decision to pursue specific product or design concepts, and to commit the resources required to mature technology and/or reduce any risks that must be mitigated prior to decisions committing the resources needed for development leading to production and fielding.

(Added-ACC) Mission Design Series (MDS)—The MDS designator is an official DoD recognized alpha-numeric designation of a military defense aerospace vehicle. A standardized set of symbols, established by the Secretary of Defense, are used for solely designating military defense aerospace vehicles categorized as: aircraft (standard or non-standard), guided missiles, rockets, probes, boosters, and satellites. The designator describes the aerospace vehicle in two components where the components are separated by a dash. The first component, comprised only of alpha characters, describes the mission of the vehicle. The second component, comprised of alpha-numeric characters, describes the design number and design series of the vehicle.

Examples include F-35A, F-35B, F-35C, F-15E, F-15EX, AIM-9L, AIM-9M.

Mission system training device—A device that provides the trainees with a simulated warfare environment that is specifically mission oriented to the type of prime mission system involved. The device can provide specific prime mission system operator modes or a mission mode that requires tactical decision-making.

Mission-task analysis—A process of reviewing mission requirements, developing collective task statements, and arranging the collective tasks in a hierarchical relationship.

Objectives/Media analysis—A process designed to identify all training objectives. It also allocates and justifies instructional strategies, methods, and media for each training objective. The objectives/media analysis defines training objectives in terms of conditions, required behavior, and standards of acceptable performance. Defines a media analysis and selection process. Documents the method/media trade process. Selects the method/media to be used and, with rationale, allocates the training objectives.

Operational training infrastructure—A framework that includes training systems, live ranges, training devices, environment generators, threat emitters, aggressors, networks, training centers, and multi-domain command and control training systems.

Operational training system—A systematically developed curriculum including, but not necessarily limited to, courseware, classroom aids, training devices, operational equipment, embedded training capability, and personnel to operate, maintain, or employ a system. The operational training system includes all necessary elements of logistic support.

Operational training—Mission-oriented training in support of operational forces readiness. It is a crucial element of all the Air Force Service Core Functions. It usually distinguishes itself from initial skills training due to its focus on employment of a weapon system and skills in an operational setting as opposed to learning the basic use of equipment or development of basic skills. Spans all domains and large scale (Tier 1) through small scale (Tier 4) events across the breadth of multiple security levels and releasability restrictions.

Orphan training system—Training devices/systems that are not supported by specific prime mission system programs or part of existing training maintenance contracts.

Prime mission system (equipment)—Any weapon system, support system, workstation, or end-item that supports a specific military mission, therefore requiring operations, maintenance, or support personnel training. Also called a defense system or parent system.

Program manager—The designated individual with responsibility for, and authority to accomplish, program objectives for development, production, and sustainment to meet the user's operational needs. The program manager has life cycle responsibility for the prime mission system. Program manager duties include providing assessments of program status and risk to higher authorities and to the operator or operator's representative; actively managing within approved resources, program cost, performance, and schedule; and providing assessments of contractor performance. Program managers report to SAF/AQ through the appropriate program executive officer (even though they may be physically located within AFMC).

Ready aircrew program—MAJCOM program that defines the minimum required mix of annual sorties, simulator missions, and training events aircrew must accomplish to sustain mission readiness.

Simulation—A method for implementing computer generated entities over time. Also, a technique for testing, analysis, or training in which real-world systems are used, or where real-world and conceptual systems are prepared by a model.

Simulator certification—The process of ensuring through validation of hardware and software baselines, that a training system and its components provide accurate and credible training. The process also ensures the device continues to perform to the delivered specifications, performance criteria, and configuration levels. It also sets-up an audit trail regarding specification and baseline data for compliance and subsequent contract solicitation or device modification.

Simulator validation—The process for (1) comparing a training device's operating parameters and performance to the current intelligence assessment of a prime mission system and threat and interaction between the prime mission system and threat; and (2) documenting the differences and impacts between the prime mission system's performance and the threat environment. This process includes generation and deployment of an intelligence data baseline of the system, comparison of simulator characteristics and performance, support for the modification and upgrade of the simulator, a comparison of simulator and threat operating procedures, and correction of any significant deficiencies. Uncorrected deficiencies are identified and published in validation reports. The process continues throughout the life cycle of the simulator.

Simulator—A training device that permits development and practice of the necessary skills for accomplishing tasks to a prescribed standard of competency in a specific prime mission system and duty position.

Synthetic environment certification—The process of ensuring, through validation of software baselines, physics-based attributes, constructive entities, and threat representation, that a synthetic environment and its components provide accurate, credible, and operationally relevant interactions to support training objectives. The process also ensures the environment continues to perform to the delivered specifications, performance criteria, and configuration levels. It also sets up an audit trail regarding specification and baseline data for compliance and subsequent contract solicitation or environment modification.

Synthetic environment—The integrated set of data elements that define the environment within which a given simulation application operates. The data elements include information about the initial and subsequent states of the terrain including cultural features, and atmospheric and oceanographic environments throughout an exercise. The data elements include databases of externally observable information about instantiable entities, and are adequately correlated for the type of exercise to be performed. More generally, it is the combination of simulators and constructive environments.

System training plan—An iterative planning document that defines the justification, design, development, funding, resources, support, modification, operation, and management of an operational training system. The STP is designed to provide for planning and implementation of training and to make sure all resources and supporting actions required for establishment and support are considered. For fielded systems, the STP may be a stand-alone document, or it may be referenced and summarized in the life cycle sustainment plan or other human systems integration documents. All references to the STP in this AFI incorporate the possibility that the intended documentation may be part of a life cycle sustainment plan or human systems integration documents. Training considerations prior to fielding will be addressed in coordination with the program manager.

Training device—A device that permits learning, development, and the practice of skills and procedures necessary for understanding and operating the integrated systems of a specific prime mission system (Simulator is a subset of Training Device). For AETC, a training device that permits development and practice of the basic skills and tasks for completion of undergraduate aircrew training to a prescribed standard of competency.

Training pipeline managers—Responsible for life cycle management (planning, directing, implementing, and overseeing) formal training courses. This process begins with basic military training and extends through advanced skills courses. Training pipeline managers administer and execute MAJCOM level training management to include prioritizing and advocating training resource requirements to the Air Staff.

Training planning team—An action group composed of representatives from all pertinent functional areas, disciplines, and interests involved in the life cycle design, development, acquisition, support, modification, funding, and management of a specific prime mission operational training system. The TPT uses the STP to ensure training considerations are adequately addressed in the prime mission system acquisition and modification processes.

Training requirements analysis—A process that allows for determination of training requirements and allocation of proficiency level to resolve a performance deficiency.

Training requirement—The knowledge, skills, and attitudes that are required for satisfying the job performance requirements and are not already in the incoming students' repertoire.

Training system basis analysis—A process that analyzes the existing operational training systems, training deficiencies, assess new training technology for potential application, evaluates alternative training system concepts, system configurations, and recommend solutions.

Training system manager—Expert who works with USAF and industry training system managers and SMEs to ensure training systems are adequately funded and properly provisioned to support USAF operations training.

Training system requirements analysis—A systematic approach to assess prime mission systems based on the instructional system development process that develops data items to document the training and preliminary system requirements to ensure personnel are taught in a cost-efficient way based on the knowledge, skills, and attitudes essential for successful mission performance. During this process, analysts extract and allocate training requirements identified from the user’s operational needs. The training system requirements analysis process is the written record from a compilation of the following four complementary elements: mission/task analysis, training requirements analysis, objectives/media analysis, and the training system basis analysis. It serves as a required input to the STP. AFLCMC, through the training system program office, acquisition program office, or AETC, and in coordination with the lead command, shall conduct the training system requirements analysis. Expanded information on the TSRA process is contained in **Attachment 2** of this AFI.

Training system support center—Normally a government-owned, contractor-operated consolidated function that includes the contractor personnel, government owned equipment, facilities, tools, and data necessary to provide configuration management functions, engineering development, and feasibility studies, life cycle hardware, software, and courseware support for an operational training system. Contractor logistics support contracted TSSCs maintain operational training system operational training baselines by merging configuration management baselined

weapon system(s) functionality with mission and training profiles, etc., to provide the using unit with an effective trainer for initial and continuation training, ready aircrew program, and distributed training capabilities. Contractor logistics support TSSCs normally support aircrew and maintenance training systems.

Training systems product group—An umbrella term to describe the USAF organizations to include AFRL and AFLCMC that support warfighter and multinational partner training through the use of modeling and simulation techniques. Collectively, the training systems product group provides the warfighter with a full range of operational training system life cycle support. Key capabilities include research and development; acquiring and/or sustaining training systems; conducting contract training support to include TSRAs, student instruction, courseware development and maintenance, training management systems, device operation and maintenance, and on-site or on-call service; assistance; concurrency modifications; technology insertion upgrades; multiple database sources; contractual vehicles for task orders; and disposition.

Training—Instruction and applied exercises for the acquisition and retention of knowledge, skills, and attitudes required to accomplish military tasks.

Using command—Any command or organization (above the Wing level) that possesses a prime mission system and uses the products of the operational training system. The using command is responsible for managing and conducting mission operations using the resources allocated by the lead command and higher headquarters. As such, the using command is responsible for defining the system requirements necessary to conduct and sustain operations. These requirements are submitted to the lead command for advocacy, programming, and funding allocation. If only one MAJCOM or agency possesses the prime mission system, that MAJCOM or agency is the designated lead command.

Utilization rates—the rate of actual resource use versus planned resource use.

Validation—(1) For the purposes of this instruction, determination of the degree to which the operational training system represents the real-world asset from the perspective of the specific training to be accomplished. The lead command is the validation authority and will designate a validation agent. (2) The process of determining the degree to which a model is an accurate representation of the real-world from the perspective of the intended uses of the model.

Verification—(1) For the purposes of this instruction, determination if the operational training system accurately represents the contract design specification or Statement of Work or Performance Work Statement. The acquisition agent is the verification authority with support from the lead command. (2) The process of determining that a model implementation accurately represents the developer's conceptual description and specifications.

Attachment 2

TRAINING SYSTEMS REQUIREMENTS ANALYSIS

A2.1. Training system requirements analysis process. The TPT will initiate a front-end TSRA after Acquisition Milestone A. Guidance covering front-end TSRAs is provided in DoDI 5000.02, *Operation of the Adaptive Acquisition Framework*, to inform STP development. The TPT will initiate and define the follow-on TSRA scope relative to the information required. **(T-1)** For new and emerging prime mission systems, the lead command will conduct a TSRA, in coordination with the training system program office, acquisition program office, using commands and/or AETC. **(T-1)** For existing systems, the lead command, program manager, or TPT will conduct recurring TSRAs when major modifications occur, including but not limited to aircraft software releases, new mission capabilities (weapons, sensors, communications), or changes to mission essential and training task lists. **(T-1)**

A2.2. Training system requirements analysis components. A TSRA is comprised of four sequential components to ensure training considerations are addressed in the prime mission system acquisition and modification processes. The TPT may direct separate front-end TSRAs to address specific mission areas, such as separate operations and maintenance TSRAs. The analytic components remain the same. The TPT may focus follow-on TSRAs to fewer components when appropriate for the defined scope.

A2.2.1. Mission-task analysis. The mission-task analysis identifies and analyzes all tasks required for the operation, maintenance, and/or support of the prime mission system. The mission-task analysis will result in a Master Task List detailing tasks for each mission. The master task list should be derived from analysis of mission tasks, associated system tasks, legacy mission systems, similar mission systems, or operational training system task lists. The TPT may provide guidance on master task list content. Based on the mission area(s) assessed, the mission-task analysis may:

A2.2.1.1. Break each mission into tasks, situational context, and coordination requirements. Each mission should be described in terms of mission objectives, scenarios, segments, and mission profiles.

A2.2.1.2. Provide a full range of threat and environmental conditions.

A2.2.2. Training requirements analysis. The training requirements analysis will develop the master task list into the training task list for the prime mission system. **(T-1)** The analysis will identify tasks requiring formal training and the criteria for successful performance in a mission context. **(T-1)** The training requirements analysis will:

A2.2.2.1. Identify target populations and tasks for which current students/trainees lack the knowledge, skills, and attitudes required for their performance. **(T-1)** Knowledge, skills, and attitudes are classified as perceptual, motor, cognitive, information processing abilities.

A2.2.2.2. Define formal training entry level and exit level knowledge, skills, and attitudes for each unique target student population. **(T-1)** Baseline target populations can include upgrade, qualification, or continuation training.

A2.2.3. Objectives/media analysis. The objectives/media analysis develops training objectives from the training requirements analysis-developed training task list and selects

delivery media. The analysis will result in clearly stated and organized training objectives mapped to an effective and feasible media solution. The objectives/media analysis will:

A2.2.3.1. Develop enabling objectives for each task defined in terms of conditions, required behavior, and lead command defined standards of acceptable performance. **(T-1)** These objectives should be organized and grouped logically under terminal objectives to use later for syllabus and course map development.

A2.2.3.2. Define and document a media analysis and selection process. **(T-1)**

A2.2.4. Training system basis analysis. The training systems basis analysis develops the training system concept and defines the training system configuration. **(T-1)** The analysis will result in a functional baseline for the design, development, and operation of an integrated training system. The training systems basis analysis may:

A2.2.4.1. Analyze legacy or similar training systems and identify deficiencies.

A2.2.4.2. Assess new training technology for potential application in training system concepts.

A2.2.4.3. Evaluate alternative training system concepts and system configurations.

A2.2.4.4. Recommend numbers, functions, and types of training media, courseware requirements, and training management system functions.

A2.2.4.5. Provide rationale and justification concerning how a proposed system will remedy deficiencies.

Attachment 3

SYSTEM TRAINING PLAN

A3.1. The Lead Command will:

A3.1.1. Establish operational training system definition through acquisition and modification documentation that will support the review and decision process. **(T-2)**

A3.1.2. Identify training needs (including networking capabilities if applicable), concepts, strategies, constraints, risks, data, alternatives, resources, responsibilities, and other areas, through an iterative process. **(T-1)**

A3.1.3. Include an instructional system development analysis of the ground-based media. Analyze how it complements hands-on training or supplements training when resource availability, security, cost, and environmental, safety, and occupational health (ESOH) constraints limit use of the prime mission system equipment as a training media. **(T-1)**

A3.1.4. Identify tasks for which personnel cannot currently be adequately trained (training gaps). These tasks will be documented in the STP as unmet requirements and identified as potential limiting factors in the ability to accomplish the prime mission system mission. **(T-1)**

A3.1.5. Document the results of early, front-end, and follow-on TSRAs. **(T-1)**

A3.1.6. Provide the basic concepts and strategy to attain and maintain operational training system concurrency and networking capability to support desired training capability at the appropriate time. **(T-1)**

A3.1.7. Establish milestones and schedules to ensure timely development, testing, and fielding of training capability and training support. **(T-1)**

A3.1.8. Identify how subsystems and components should be integrated into the total operational training system. Include distributed training capabilities (for example, DMO) implementation, sustainment, and utilization, if applicable. **(T-1)**

A3.1.9. Provide information and identify resources for management decisions within the planning, programming, budgeting, and execution system process which support operational training system acquisition, modification, and sustainment processes. **(T-1)**

A3.1.10. Identify alternate training strategies, to include methodology and media, if funding, concurrency, or other unknowns negatively impact required operational training system capabilities. **(T-1)**

A3.1.11. Recommend areas for new technology applications to improve future operational training system effectiveness and efficiency. **(T-1)**

A3.2. System Training Plan format. The TPT may choose to utilize one of two formats for the STP depending upon the life cycle phase of the prime mission systems: emerging mission systems and existing mission systems.

A3.2.1. STP format for emerging prime mission systems. The exact composition of the STP is at the discretion of the TPT. The STP will be referenced and summarized in the life cycle sustainment plan or other human systems integration documents. The STP is a top-level document that provides input to the requirements generation, acquisition program planning,

and budget development processes. The details needed to support this analysis may be maintained in other documents and referenced in the STP.

A3.2.2. Suggested STP sections. Include only those sections necessary to guide the development, fielding, and management of the operational training system:

A3.2.2.1. Executive summary. Provide an overview of the STP. Highlight sufficient and significant elements to support the program, shortfalls, and future objectives. Briefly describe the overall mission of the prime mission system, the operational training system, and requirements. Show the relationship of the resource to meeting the overall mission, shortfalls, and alternatives.

A3.2.2.2. Mission and prime mission system description. Describe the mission and prime mission system based on the operational requirement, threat environment, and the designed operational capability, when determined. Include a thorough analysis of the mission performed by the prime mission system; a classified attachment may be required. Include title, nomenclature, and program elements for budget, security classification, prime mission system priority rating, and principal agencies. Reference other plans and documents that support the prime mission system mission or operational training system acquisition and modification process. Include a brief summary of the baseline system to be replaced, modified, or augmented; shortcomings, displacement, or disposition, if being replaced.

A3.2.2.3. Training planning team (TPT) membership. The STP should document TPT membership, which must comprise the lead command, using commands, weapon system program office, and training systems product group representatives.

A3.2.2.4. Operational training system description. Describe the total operational training system by functional area, including instructional strategy, duration, content, media, training devices and utilization rates, and facilities. Provide strategy and alternative methodologies throughout the training continuum. Address the use of distributed training capabilities (for example, DMO) or describe how this requirement will be waived in coordination with using commands and the training systems product group. Identify proposed approach to acquire training equipment and facilities. Estimate training qualification time required to achieve full proficiency. Include description of database, systems integration, networking standards, controlled interfaces, compatibility, transportability, and deployment requirements. Address ability to efficiently and cost effectively modify operational training system software concurrently with the prime mission system. Identify requirement for computer-based training and interactive courseware. Provide a course summary document.

A3.2.2.5. Diagram a timeline which includes the training progression of each operational and maintenance functional area from entry into and through exit from the prime mission system. Identify on the continuum all qualification levels, evaluation checkpoints, and reentry qualification points. State the policy upon which decisions will be based for critical points, such as course sequence and media allocation on prime mission system equipment training. Indicate basic training principles to be considered, such as a building-block approach.

A3.2.2.6. Describe operational training systems.

A3.2.2.7. List and describe operational training system component's role, use, and capabilities:

A3.2.2.7.1. Actual prime mission system mission and non-prime mission system equipment.

A3.2.2.7.2. Courseware and associated equipment.

A3.2.2.7.3. Training aids and devices.

A3.2.2.7.4. Embedded training capability in the weapon system.

A3.2.2.8. Describe AFRC and ANG participation.

A3.2.2.9. Identify all composite, joint, interagency, or multinational training applications.

A3.2.2.10. Address potential or unresolved training issues.

A3.2.2.10.1. Operational training system requirements. Summarize the key capability needs and operational and sustainment requirements for the operational training system as expressed in appropriate operational capability documents, concepts of operations, and other basic requirements documents. Describe how manpower, personnel, training, human factors engineering, safety, and occupational health considerations are applied to the design and development of the prime mission system operational training systems to reduce costs and enhance capabilities. Establish initial objectives that support readiness, force structure, affordability, and operational objectives.

A3.2.2.10.2. Training system requirements analysis (TSRA). The TPT will use the results of the TSRA to identify the prime mission system training requirements. The TPT will validate the TSRA products for use in the design of the operational training system. The mission-task analysis report and training requirements analysis report will be used by the TPT to develop the training task list with performance criteria for inclusion in the operational training system requirement documents. The objectives/media analysis report and training system basis analysis report may be used to identify other operational training system requirements, such as the numbers and types of training devices, courseware, potential training alternatives, technology assessment, observations, and recommendations to be incorporated into the training requirement documents.

A3.2.2.11. Implementation. Describe data sources, implementation procedures, special authorization or approvals, and assign responsibilities. Identify those training areas not supported by a complete task analysis process. Describe the use of distributed training capabilities (for example, DMO) or how this requirement will be waived in coordination with using commands or training systems product group.

A3.2.2.12. Operational training system concurrency strategy. Identify and group critical training tasks consistent with mission training development and implementation that are impacted by concurrency. When incremental (phased) delivery of training capability is advantageous or necessary, training capabilities should support the following priorities as agreed to by the TPT:

A3.2.2.12.1. Safety training requirements and tasks.

A3.2.2.12.2. Warfighting training requirements and tasks.

A3.2.2.12.3. Full mission training and rehearsal requirements and tasks.

A3.2.2.13. Organizational interfaces. Identify government organizations necessary to ensure timely approvals and transfer of data, equipment, and property, which should be concurrent with the first contract award and renewed throughout the life cycle of the prime mission system operational training system. Include established agreements such as service-level agreements, statements of work, and memorandums of understanding. Briefly list responsibilities for each command or organization.

A3.2.2.14. Operational training system management and support concept. Concurrency must be given primary consideration in contracting. Identify the concept and strategy for achieving life cycle management and support of the operational training system, to include cybersecurity. Describe requirements and options for logistics support. Contractor logistics support contracts that include hardware and/or software modifications should be developed and used. Consider:

A3.2.2.14.1. Contractor logistic support management.

A3.2.2.14.2. Technical data.

A3.2.2.14.3. Spares.

A3.2.2.14.4. Consumables.

A3.2.2.14.5. Organizational-, intermediate-, and depot-level maintenance.

A3.2.2.14.6. Special or system operational equipment.

A3.2.2.14.7. Common or special tools and equipment.

A3.2.2.14.8. Facilities.

A3.2.2.15. Manpower support concept, military personnel utilization concept, and personnel training requirements. Consider student demographics, entry requirements, and student throughput estimates; estimate portion of military, civilian, or contract personnel, to include Cybersecurity requirements. Describe Air Force Specialty Code employed. Identify these and any other unique requirements for this system in each of the following functional areas:

A3.2.2.15.1. Combined test force.

A3.2.2.15.2. Initial cadre.

A3.2.2.15.3. Operations.

A3.2.2.15.4. Maintenance.

A3.2.2.15.5. Depot.

A3.2.2.15.6. Security forces.

A3.2.2.15.7. Munitions and explosive ordnance.

A3.2.2.15.8. Contract support, contractor logistics support, or contract training.

A3.2.2.16. Training Constraints and Risks. Include all potential limitations that will or may affect timely implementation of training objectives to meet initial operational capability and maintain full operational capability. Describe all peacetime training

constraints. Consider manpower or personnel and resource availability, security, cost, and ESOH considerations, which may influence training media and methodology design, development, and selection. Program managers document the status of operational training system (or any other type of system) ESOH risk management in the Programmatic Environment, Safety, and Occupational Health Evaluation (PESHE). The lead command and using command should refer to the PESHE for the identification and status of ESOH risks. For operational training system programs, the ESOH risk acceptance authorities should be aligned with DoDI 5000.02. Include peacetime restrictions on the use of the prime mission system. Identify risks and assign risk levels that may affect deployment schedules or other milestones. Identify the expected impact of late to need or unusable training devices in terms of workarounds, dollar costs for alternative training, increased use of the prime mission system, or impact of failure to perform on combat capability. Consider initiatives such as advanced prime mission system design change data deliveries and long-lead contractor provided equipment or government-furnished equipment, information, or property. The risk government-furnished property adds to a concurrent delivery schedule must be discussed and tradeoffs identified.

A3.2.2.17. Prime mission system and operational training system milestones. Identify the prime mission system and operational training system schedules and priority ratings necessary for concurrency required to deliver the operational training system. Show "need dates" in terms of milestones. Include key engineering change proposals, management responsibility, and operational milestones. Consider all schedules pertinent to satisfying training objectives through definitive milestones. These could include:

A3.2.2.17.1. Weapon system major milestones to include initial operational capability through full operational capability.

A3.2.2.17.2. Task requirements and analyses completion dates.

A3.2.2.17.3. Training equipment requirements and delivery.

A3.2.2.17.4. Facility beneficial occupancy dates.

A3.2.2.17.5. Prime mission system and operational training system deployment dates.

A3.2.2.17.6. TSSC activation.

A3.2.2.17.7. Factory or contractor training dates.

A3.2.2.17.8. Instructional course start dates.

A3.2.2.17.9. Logistics support requirements dates.

A3.2.2.17.10. Ready for Training and Required Assets Available dates.

A3.2.2.17.11. Technical data availability.

A3.2.2.17.12. Courseware development completion dates.

A3.2.2.17.13. Training management system completion dates.

A3.2.2.17.14. Operational training system evaluation plan and review dates.

A3.2.2.17.15. Distributed training capability (for example, DMO) implementation and sustainment plan, training objectives, initial operational capability, and full operational capability dates.

A3.2.2.18. Resource Summary. Identify total resource requirements to develop and operate the operational training system throughout the prime mission system life cycle. Include recommended tradeoffs to support training and impact of not funding or procuring desired training capability.

A3.2.2.18.1. Indicate funding by allocation and fiscal year.

A3.2.2.18.2. Include training or test equipment, courseware, training aids, technical manuals, and documentation by types, numbers, and life cycle support.

A3.2.2.18.3. Manpower, to include officer, enlisted, and civilian.

A3.2.2.18.4. Personnel, to include instructor cadre and support personnel.

A3.2.2.18.5. Military construction or facility modification. Describe project and costing by fiscal year. Establish physical, power, security, and other relevant requirements.

A3.2.2.18.6. Contractor support. Time, effort, and cost. Initial training support.

A3.2.2.18.7. Travel and per diem requirements and costs.

A3.2.2.18.8. Other: Airspace, ranges, flying hours, munitions, etc.

A3.2.2.19. Training evaluation and validation. Develop and document evaluation and validation criteria, methodology, and responsibilities. Provide cost-benefit analysis of proposed alternatives. Include plan for evaluation of training effectiveness.

A3.2.2.20. Research and development efforts. Describe current and future research and development studies and cost-benefit analysis that may support upgrades to the systems or alternative methodologies to close any training gaps or accomplish the training with fewer resources.

A3.2.2.21. Lessons learned. Identify problem areas common with other programs and potential solutions. Document assumptions made, fixes, workarounds, or changes to requirements based on lessons learned. Include impact on system costs, effectiveness, and combat capability.

A3.2.2.22. Distribution. Distribute to members of the TPT and other designated agencies.

A3.2.3. STP format for existing prime mission systems. After fielding of the operational training system, the STP for an emerging weapon system becomes a historical document providing direction, perspective, and guidance for managers of the operational training system. The STP for an existing weapon system is a road map of the operational training system. The exact composition of the STP is at the discretion of the TPT. It may include the following:

A3.2.3.1. An assessment of future training needs caused by changes in the prime mission system and/or its mission tasking.

A3.2.3.2. A timeline to show the plan for sustaining, modifying, disposing, and replacing the operational training system components.

A3.2.3.3. Any analysis, assessment, or background documentation that provides justification for acquisition, modification, and funding support for operational training system components. Include distributed training capabilities (for example, DMO) implementation and sustainment, if previously included.

A3.2.3.4. An assessment of operational training system deficiencies and their impact on the operational training system costs, effectiveness, and combat capability. Document recommended fixes, work-a-rounds, or changes to requirements.

A3.2.3.5. An assessment of future research and development efforts or technological advances that could improve training effectiveness and/or efficiency, including cost-benefit analysis data. Include distributed training capability, timelines, and funding considerations.

A3.2.3.6. A current version of the PESHE.

A3.3. (Added-ACC) Functional Area Managers (FAM). FAMs should provide their latest STP to their Training Planning Team for updates and signatures, as needed. The STP can then be used to support funding requests for needed training modifications/upgrade(s). **(T-2)**

A3.4. (Added-ACC) Updates. Updates to the STP should be in place prior to any major modification(s)/upgrade(s).

A3.5. (Added-ACC) Current STP. The latest STPs should be kept on digital file in the A3 branch responsible for the MWS. A copy should be sent to ACC/A3TO for central storage.

Attachment 4

TRAINING SYSTEM METRICS

A4.1. Training System Metrics Tracking. Training system metrics are used to track training system simulator utilization and to communicate the status and health of training systems through the MAJCOMs to AF/A3TI. The intent is to gain and maintain visibility on issues affecting training systems and measure and improve on how simulator training is conducted. MAJCOMs will track, as a minimum, the metrics data shown in **Figure A4.1**. **(T-1)** Utilization will be tracked for simulators that are used to meet AFI11-2MDS Vol 1 currency and Ready Aircrew Program requirements and for simulators that use contracted hours to provide contractor instructor-based training. **(T-1)** Simulators that have contractor logistics support but do not contract for set availability hours will report "Not Applicable" for contracted hours. **(T-1)** IAW **paragraph 2.12.2.10**, simulator utilization metrics will be reported to AF/A3TI semi-annually. Report by 30 April for the prior semi-annual period October through March and by 31 October for the prior semi-annual period April through September. MAJCOMs will upload their data to the platform-specific worksheets located in their respective folders in Synthetic Training and Test folder on the OTTI Capability Development Team sharepoint site: **https://usaf.dps.mil/sites/HAF-A5/center2/otti_cft/SitePages/OTTI.aspx**. **(T-1)** Forward any questions to the AF/A3TI Workflow email address (**AF.A3TI.Workflow@us.af.mil**).

Figure A4.1. Training System Simulator Metrics.

Report Month and Year:										
Contract ID:						**PEC:**				
Type of Contract:						**Cost Cat:**				
(A) Prime Mission System	MAJCOM	Base / Location	Device Serial Number	(B) Type of Training Device	(C) Available Hours Contracted for Use	(D) Official Purpose Hours	(E) Hours Executed	Utilization Rate: E divided by D	(F) Official Purpose Hours Not Used: D minus E	Hours used for DMO
Total										

A4.2. See Figure A4.1 column headers explained below:

A4.2.1. {A} Prime Mission System—See **attachment 1** Terms (i.e., F-15C, C-130J, JTAC, AOC).

A4.2.2. {B} Type of Training Device—Nomenclature for the training system device (i.e., WST, OFT, FMT, FMS, PPT, CPT, MCTS).

A4.2.3. {C} Available Hours Contracted for Use—Hours the contractor is required to have the training device available to be scheduled for Official Purpose use.

A4.2.4. {D} Official Purpose Hours—Contracted hours used for Official Purposes (See **paragraph A4.3**.).

A4.2.5. {E} Hours Executed—Official Purposes Hours actually used.

A4.2.6. {F} Official Purpose Hours Not Used—Hours the training device was not being used for Official Purposes. Provide rationale breakout in **Figure A4.2** for utilization rates less than 80%.

A4.3. Examples of Official Purposes:

A4.3.1. Aircrew Training (Scheduled or Unscheduled).

A4.3.2. Simulator modification/upgrade/maintenance.

A4.3.3. Simulator facility/bay maintenance.

A4.3.4. Contractor use (pre-mission simulator set-up, pre-briefs, post-briefs, syllabus prep, instructor training, etc.).

A4.3.5. Very Important Person Tours.

A4.3.6. Maintenance Training.

Figure A4.2. Training System Simulator Non-Use Breakout.

Reason	Number of Official Purpose hours not used (F)
Flex time built into contract	
Unscheduled simulator repair	
Unscheduled simulator facility/bay repair	
Weather (lightning, flooding/leaking roof, heat/HVAC, etc.)	
Crew no-show (reason unknown)	
Short-notice crew cancellation for operational reasons	
(List other reasons as applicable)	
Total [Must match total of column (F) on Table A4.1]	

Attachment 5

TRAINING SYSTEM SIMULATOR AND SYNTHETIC ENVIRONMENT CERTIFICATION AND SIMULATOR VALIDATION

A5.1. General. Training system simulator/synthetic environment certification ensures that training systems and their components support accurate and credible networked (for example, DMO) and non-networked training for allocated tasks, missions, and events through training system hardware and software performance.

A5.1.1. Simulator validation verifies and validates the networked and/or non-networked simulated weapon system and self-generated mission environment performance. JSE-compliant "Aircraft In A Box" simulators will be used to validate the simulator using JSE as the synthetic environment. Simulator and environment validation will be documented separately. **(T-1)**

A5.1.2. Operational training system training device simulator/synthetic environment certification and simulator validation examine fidelity, concurrency, connectivity, and usability characteristics to support operational training system accreditation and identify capabilities and limitations.

A5.1.3. Operational assessments supporting operational training system acquisition, modification and modernization may use simulator/synthetic environment certification and simulator validation methodologies managed by the acquisition agent and supported by the lead command.

A5.1.4. Simulator/synthetic environment certifications and simulator validations will recur throughout the life cycle of the system/environment, and each assessment will focus on training fidelity and quality of a lead command-approved, TSSC-developed, hardware and software configuration for that device. **(T-1)**

A5.1.5. The reports generated by the simulator/synthetic environment certification and simulator validation support the Accreditation Authority's decision to accredit the associated operational training system for its intended use.

A5.1.5.1. **(Added-ACC)** ACC/A3 is the accreditation authority for TSs used to meet 11-2MDS Volume 1 currency and Ready Aircrew Program (RAP) Tasking Memorandum (RTM) training requirements. SIMCERT/SIMVAL or other 29 TES reports, assessment results, and recommendations should be the basis of accreditation along with inputs from OG/CCs.

A5.1.5.2. **(Added-ACC)** CAF MAJCOMs may use ACC's SIMCERT report or accreditation package for assigned TSs with the same standalone configuration training and mission baselines.

A5.2. Simulator certification program.

A5.2.1. Evaluate the simulator in an operational training environment and assess the following:

A5.2.1.1. Concurrency, connectivity, and fidelity.

A5.2.1.2. Major systems (for example, visual, motion, threat systems, CGFs, aerodynamic, weapons, sensors, electronic warfare, and control input).

A5.2.1.3. Instructor functions and interfaces.

A5.2.1.4. Brief/debrief and performance measurement.

A5.2.1.5. Integration and interoperability.

A5.2.1.6. **(Added-ACC)** Ability to conduct RAP Missions and Events, Instrument and Emergency procedures.

A5.2.1.7. **(Added-ACC)** Ability to conduct distributed training with other platforms as identified by A3 divisions.

A5.2.2. Simulator certification accreditation. The simulator certification agent will provide certification results and the accreditation recommendation to the lead command, or designated accreditation proponent, for review.

A5.2.2.1. **(Added-ACC)** 29 TES will deliver SIMCERT Report to A5T and appropriate A3 MDS FAM. **(T-2)** A3 MDS FAM will review SIMCERT Report and coordinate with OG/CCs or designated representatives to develop a TD/TS Accreditation Memorandum for A3 Signature. **(T-2) Note:** A3 signature on annual RTM ***does not*** constitute TS accreditation.

A5.2.3. **(Added-ACC)** OG/CCs may recommend a TS be decertified for some or all AFMAN 11-2MDS Volume 1 currencies or RTM Mission and Event requirements until a new SIMCERT or other OT evaluation is accomplished.

A5.3. Synthetic environment certification program (not required for self-generated mission environments integral to the simulator).

A5.3.1. Evaluate the synthetic environment in an operational training environment and assess the following:

A5.3.1.1. Physics and threat representation.

A5.3.1.2. Concurrency, connectivity, and fidelity.

A5.3.1.3. Major systems (for example, image generation, threat systems, constructive entities, observable states, red/blue weapons, sensors, electromagnetic environment, and multi-spectral environmental effects).

A5.3.1.4. Instructor functions and interfaces.

A5.3.1.5. Data capture for brief/debrief and performance measurement.

A5.3.1.6. Integration and interoperability.

A5.3.2. Synthetic environment certification accreditation. The synthetic environment certification agent will provide certification results and the accreditation recommendation to the lead command, or designated accreditation proponent, for review.

A5.3.3. Synthetic environment certification should be accomplished for both networked and non-networked activities.

A5.4. Simulator validation program.

A5.4.1. Simulator validation is the process for (1) comparing a training device's operating parameters and performance to the current intelligence assessment of a prime mission system, threat, and interaction between the prime mission system and threat; and (2) documenting the differences and impacts. This process includes generation and deployment of an intelligence data baseline of the system, comparison of simulator characteristics and performance, support for the modification and upgrade of the simulator, a comparison of simulator and threat operating procedures, and correction of any significant deficiencies. Uncorrected deficiencies are identified and published in validation reports. The process continues throughout the life cycle of the simulator.

A5.4.1.1. **(Added-ACC)** LC is the validation proponent and the training accreditation authority for TS procured or sustained by respective LC. **(T-2) Note:** This program does not apply to maintenance training devices, whose functional performance is assessed by TO compliance verification.

A5.4.1.1.1. **(Added-ACC)** ACC SIMVAL is a significantly more detailed technical evaluation process than SIMCERT. It is categorized as a "major system" operational assessment of the CGF mission environment and of all interfaces comprising the MWS TS. Units conducting SIMVAL require technical expertise, manning, and resources vastly different than those needed for SIMCERT. Results and recommendations are included in the SIMVAL operational report and provide the artifact for MAJCOM /A3 accreditation of the entire TS.

A5.4.1.1.2. **(Added-ACC)** 53 WG's 29 TES is ACC's overarching SIMVAL Agent. Other test expertise may be called upon with 53 WG or ACC's T&E Branch (ACC/A5TT) approval to participate in SIMVALs as appropriate in order to help assess TSs in areas such as threat entities and electronic warfare environment generators, weapon effects, etc.

A5.4.2. Operational training system simulator validation performed on a common CGF or HITL may not be used as the only source in formulating the accreditation decision for all operational training systems that use that CGF or HITL because the interactions between the simulated weapon systems and the simulated threats may not be consistent or equivalent. Regression testing must be accomplished to verify the consistency of the CGFs or HITLs interactions with the simulated weapon systems and to confirm the operational training system uses CGF or HITL inputs correctly. JSE-compliant "Aircraft In A Box" simulators will be used to certify the simulators using JSE as the synthetic environment.

A5.4.3. Simulated physical systems accurately representing entities in all ways are needed to stimulate all platform sensors and provide needed characteristics for weapons employment and primary training activities. The emphasis will be on assuring modeling to appropriate fidelity for the operational training system.

A5.4.4. New threats or changed threats in the CGF or HITL entity tables may require a simulator validation.

A5.4.5. Operational training system simulator validation acquisition of systems and modifications programs will be conducted IAW AFI 16-1001 and MAJCOM guidance. The simulator validation will include:

A5.4.5.1. Physical models. Validate physical performance and environmental interaction models in the CGF with authoritative sources: (for example, intelligence community estimates, test results, instrumented range data, etc.).

A5.4.5.2. Behavioral models. Base individual and organizational behavior models on accepted cognitive modeling methodologies. They will be executed at fidelity levels consistent with the warfighter tactical interactions.

A5.4.5.3. A summary of operational training system differences from threat and live weapon systems.

A5.4.5.4. Simulator validation requirements in the STP, life cycle sustainment plan, or human systems integration.

A5.4.6. Operational training system simulator validation accreditation. The operational training system simulator validation and accreditation results are forwarded to appropriate MAJCOM and unit leadership.

A5.4.6.1. **(Added-ACC)** CAF MAJCOMs may use ACC's SIMVAL accreditation package for assigned TSs with the same standalone configuration training and mission environment baseline.

Attachment 6

THREAT MATRIX FRAMEWORK (TMF)

A6.1. Overview. TMF is intended to be used in planning and execution for effective training event mission accomplishment. It provides a standardized assessment framework for air, space, and cyberspace domains and is designed for tactical applications for unit training. TMF is not designed to measure the capabilities of an entire country/region, but only smaller vignettes and is not intended to replace in-depth analysis that requires an analyst in the loop. To apply this standard framework to a scenario, one must bound the scenario by geography, time, and political situation. There are inherent limitations underlying the TMF. When applied to an enemy in a scenario, the TMF is an estimate of that enemy. This approximation is not intended to replace the in-depth analysis behind a TMF assessment but act as an augmentation to enable better communication across differing MDS or communities. For example, an environment labeled as contested could actually be permissive for a 5th generation aircraft yet simultaneously be highly contested for a C-17. Utilizing the TMF and numerically categorizing the adversary threat environment overcame the ambiguity associated with terms such as contested and permissive. A level-2 fighter is a level-2 fighter threat to every MDS.

A6.2. TMF contains criteria in each of the following nine categories:

A6.2.1. Air Surveillance.

A6.2.2. Battle Management.

A6.2.3. Fighters.

A6.2.4. Intercept Control.

A6.2.5. Surface Based Air Defense.

A6.2.6. Electronic Attack.

A6.2.7. Cyber.

A6.2.8. Space.

A6.2.9. Integration.

A6.3. These categories list the criteria and what is required (“Yes/No”) for each threat level. To aid in discriminating between the various Threat Levels these criteria make the following distinctions:

A6.3.1. "No" simply means it is not "Required" to Train/replicate a certain Threat Level, it may still be highly "desired" at a Lower Threat Level.

A6.3.2. Threat replication type and density is dependent on the TMF Level; while every attempt was made to create specific details for each TMF Level, functional area manager analysis is needed to discern a Level 3 vs 4 "advanced threat".

A6.3.3. The assessment is based on operational training requirements for CMR aircrew/operators not Undergraduate or Initial Qualification criteria.

A6.4. TMF Level Definitions.

A6.4.1. TMF 0-2—Defined as the ability to replicate and train against varying degrees of lower threat environments.

A6.4.2. TMF 3—Defined as the ability to replicate and train against an advanced adversary with legacy capabilities.

A6.4.3. TMF 4—Defined as the ability to replicate and train against a peer/near-peer adversary.

A6.4.4. TMF 5—Defined as the ability to replicate and train against emerging/future capabilities.

Attachment 7 (Added-ACC)

GOVERNMENT CONTRACT OVERSIGHT

A7.1. (Added-ACC) Contracting Officer's Representative (COR). The COR is the sole government representative on site. As such, CORs serve as technical managers assessing contractor performance against contract performance standards as authorized in the contracting officer's delegation package. The JPO has the responsibility to provide CORs for JPO contracted TSs and operations.

A7.1.1. **(Added-ACC)** The Contracting Officer will provide an LOD to requiring activity (i.e., ACC units) COR. **(T-2)** The LOD specifies COR contract oversight and surveillance responsibilities, authority, and limitations of authority.

A7.2. (Added-ACC) The Contracting Officer's Representative (COR):

A7.2.1. **(Added-ACC)** monitors operational readiness/availability of TS operations and scheduling office for impacts to availability.

A7.2.2. **(Added-ACC)** reviews AFTO Forms 781A, *Maintenance Discrepancy and Work Document,* and AFTO Form 244, *Industrial/Support Equipment Record,* IAW the Quality Assurance Surveillance Plan (QASP) and report all discrepancies delayed over 30 days, including repeat discrepancies for inclusion in the monthly contractor's performance evaluation reports.

A7.2.3. **(Added-ACC)** monitors technical data IAW the QASP and documents contractor performance as appropriate.

A7.3. (Added-ACC) Requiring Activity (Unit) TS COR.

A7.3.1. **(Added-ACC)** The Unit will nominate and provide a COR for TS CLS contracts. The Unit COR will complete the nomination process using the automated tools identified by the contracting officer. **(T-2)** Units must ensure nominated and appointed CORs are able to dedicate sufficient time to perform effective oversight of the CLS contractor as delegated by the contracting officer. **(T-2)**

A7.3.2. **(Added-ACC)** The Unit COR is the on-site primary point of contact between government organizations and the TS contractor. As such, CORs coordinate all government administrative, logistic, and security requirements as required by the contract. The Unit COR will:

A7.3.2.1. **(Added-ACC)** monitor contractor performance and provide monthly reports IAW the contracting officer's LOD. CORs will also send a courtesy copy of their reports to the appropriate A3 and A5 divisions as requested. **(T-2)**

A7.3.2.2. **(Added-ACC)** coordinate outside program organization requests for TS utilization or facility access. CORs also ensure security clearances are processed IAW established unit procedures.

A7.3.2.3. **(Added-ACC)** monitor ordering and use of government furnished equipment.

A7.3.2.4. **(Added-ACC)** monitor, verify, and report TS contractual availability and utilization and submit monthly reports IAW **Attachment 5**.

A7.3.2.5. **(Added-ACC)** assist in development of TS Requirement Documents.

A7.3.2.6. **(Added-ACC)** review and validate contractor Field Service Reports.

A7.3.2.7. **(Added-ACC)** notify the MAJCOM OPR and the contracting officer when assigned TS is not available for training due to an unplanned shutdown lasting for more than 24 hours.

A7.3.2.8. **(Added-ACC)** develop and maintain QSAP for assigned TS contracts to monitor, document, evaluate, and report contractor performance IAW the contracting officer's delegation.

A7.4. (Added-ACC) Requiring Activity (Unit) TSSC COR.

A7.4.1. **(Added-ACC)** TSSCs support all units (throughout the CAF) with TSs of the same MWS configuration. This action is typically performed through collaboration among the contracting officer, Defense Contract Management Agency, and the host unit.

A7.4.2. **(Added-ACC)** When delegated by the contracting officer, CORs that surveil and oversee TSSCs are also responsible for managing TSSC workload as identified in the contract and LOD. CORs assigned to TSSC contracts should not be tasked with other unit-level duties to ensure the COR is available to dedicate sufficient time to perform effective oversight and management of the TSSC. This COR also performs as the point of contact between government agencies and the TSSC support contractor. Furthermore, the COR facilitates TSSC administrative, logistic, and security requirements.

A7.4.2.1. **(Added-ACC)** For training systems requiring support of a TSSC, a position will be provided which will be solely dedicated to COR and Site Lead (acts as point of contact between government agencies and the TSSC support contractor; facilitates TSSC administrative, logistic, and security requirements) duties. This position will be funded by the platform Program Element (PE) and the Unit Manning Document duty title will read "(platform name) TSSC COR/Site Lead".

A7.4.3. **(Added-ACC)** TSSC COR manages the TSCWG's prioritized projects/tasks assigned to the TSSC contractor. The TSCWG is an LC chartered and chaired organization. See **Attachment 8**.

A7.4.4. **(Added-ACC)** TSSC COR, if assigned, develops and maintains a QASP to monitor TSSC activity and track activities associated with timely completion of TSCWG requirements and priorities. The TSSC COR also coordinates SME and/or test personnel support to evaluate modification projects for compliance with operational and contractual requirements.

A7.4.4.1. **(Added-ACC)** TSSC COR reviews and validates contractor TSSC Accomplishment Reports.

A7.4.4.2. **(Added-ACC)** TSSC COR participates in Physical Configuration Audits and Functional Configuration Audits IAW the contract and supports contract source selection, program/engineering design reviews, TIMs, integrated product teams, audits, site activation meetings, test and logistics planning working groups, and award conferences for acquisition, modification, and support of TS.

A7.4.4.3. **(Added-ACC)** TSSC COR also evaluates MWS and other deficiency reports for TS impact and presents them to the TSCWG for review and prioritization.

Attachment 8 (Added-ACC)

TRAINING DEVICES (ATD & MSTD) MODIFICATION APPROVAL PROCESS

A8.1. (Added-ACC) TD modifications: TD modifications are processed using an AF Form 1067, *Modification Proposal*. Guidance for completing and routing AF Form 1067 is included in Department of the Air Force Instruction (DAFI) 20-101/63-101, *Integrated Life Cycle Management*; *A589 Delegation of RDA and 1067 Policy* - 2 Jan 2024, and this attachment. The F-35 JPO does not rely solely on AF Form 1067 for requirements generation and validation. Modifications to F-35 training systems follow JPO processes.

A8.1.1. **(Added-ACC)** Weapons system approved AF Form 1067s are source documents for TD modifications driven by weapons system (e.g., aircraft) changes. AF Form 1067 is not required for TSSC changes that correct minor discrepancies and/or deficiencies associated with an approved modification.

A8.1.2. **(Added-ACC)** AF Form 1067s for TD unique modifications must be reviewed and approved by the System Program Office (SPO) and certified and approved by ACC as LC. TSSC accomplished capability modifications may be packaged as block updates and submitted on AF Form 1067 for LC validation prior to entering the Simulator SPO configuration control board (CCB) process. The AF Form 1067 process for TD unique modifications is separate from the weapons systems driven (e.g., aircraft) change AF Form 1067 database.

A8.2. (Added-ACC) Training System Working Group (TSWG). The TSWG is a MAJCOM user forum for identifying and reviewing training shortfalls and proposing and prioritizing modifications for the TS. The TSWG will prioritize the requirements identified by TSWG members, based on operational training inputs/requirements. A charter should be developed and maintained to document the TSWG process and membership roles and responsibilities.

A8.2.1. **(Added-ACC)** The primary purpose of the TSWG is to provide sustainment and ensure CAF training systems fidelity and concurrency for the lifecycle of the device which includes courseware, hardware, and software support. The TSWG will review and evaluate TD and TS change proposals, discrepancy reports or other TS deficiency reporting process, engineering feasibility studies, mission generation requirements, and database updates to provide recommendations for modifications. **(T-2)** The TSWG outcome (product) validates and provides prioritized training requirements and TS functionality changes for action with A5T, or Simulator Program Office (SIMSPO) contract (e.g., Engineering Change Proposal (ECP)) activity. When required, an AF Form 1067 will be initiated IAW **paragraph A8.5** below. **(T-2)**

A8.2.2. **(Added-ACC)** A3 and A5 divisions chair the TSWG for TD and TS within their area of responsibility. A3/A5 division may designate a lead wing as co-chair of the TSWG associated within their respective weapon system. The TSWG Chair (or designated representative) will coordinate and schedule meetings, provide agendas, and distribute minutes. **(T-2)** The TSWG should occur bi-annually and be held at the TSSC site.

A8.2.3. **(Added-ACC)** Other TSWG representatives include, unit operations, unit COR, Contract Officer Technical Representative, TSSC COR, program office, contractor, and respective MAJCOM OPR. ACC divisions, MAJCOM OPR, and unit operations

representatives should be voting members of the TSWG. All other TSWG members are advisors.

A8.3. (Added ACC) Training System Configuration Working Group (TSCWG). The TSCWG is a MAJCOM level forum for reviewing and prioritizing change proposals to the operation (training-centric) and functions of a TS. Once AF Form 1067 is generated from the TSWG products, the TSCWG determines the course on how these deficiencies will be corrected. The TSCWG outcome (product) validates and provides prioritized TS functionality changes for TSSC action or SIMPO contract (e.g., ECP) activity. A charter should be developed and maintained to document the TSCWG process and membership roles and responsibilities.

A8.3.1. **(Added-ACC)** The primary purpose of the TSWG is to review and evaluate the TSWG products, TD and TS change proposals, engineering feasibility studies, mission generation requirements, and database updates to provide recommendations and priorities for modifications. **(T-2)**

A8.3.2. **(Added-ACC)** The TSCWG will prioritize TSSC tasking, studies, and data base updates based on operational training inputs/requirements. **(T-2)** Projects that are beyond the TSSC level of effort will be processed IAW DAFI 20-101/63-101 and this attachment for possible ECP by the appropriate A5 Division Weapons System Team. **(T-2)**

A8.3.2.1. **(Added-ACC)** A3 and A5 divisions chair the TSCWG for TD and TS within their area of responsibility. A3/A5 division may designate a lead wing as co-chair of the TSCWG associated with their respective weapon system. Other TSCWG representatives include, unit Contract Officer Representative, Contract Officer Technical Representative, TSSC COR, program office, and contractor representatives. ACC divisions, MAJCOM OPR, and CORs should be voting members of the TSCWG. All other TSCWG members are advisors.

A8.3.3. **(Added-ACC)** TSCWG Chair. The TSCWG Chair (or designated representative) will coordinate and schedule meetings, provide agendas, and distribute minutes. **(T-2)** The TSCWG should occur bi-annually.

A8.4. (Added-ACC) Level of Effort. TSSC level of effort AF Forms 1067 may be required as an artifact document for the SIMSPO CCB. The appropriate ACC Division will validate and forward these AF Forms 1067 to the appropriate program office for processing. **(T-2)**

A8.5. (Added-ACC) AF Forms 1067 Preparation and Processing. Block 1 through 5, 7, and 11 through 16 are self-explanatory. Leave Part IV blank except as indicated below. **Note:** IAW DAFI 20-101/63-101, all AF Forms 1067s for new capability modifications must include (as an attachment) a Performance Parameters/Attributes Table (Key Performance Parameter, Key System Attributes, or other attributes, reference is the Joint Capabilities Integration and Development System (JCIDS) Manual). This applies to all AF Forms 1067 requiring MAJCOM or AF/A5R approval or ACC Rehearsal of Concept (ROC) validation and the Vice Chief of Staff of the Air Force approval.

A8.5.1. **(Added-ACC)** Block 8, *PURPOSE*: state in clear, concise words the deficiency to be corrected or the need to be satisfied and the expected result.

A8.5.2. **(Added-ACC)** Block 9, *IMPACT*, describe the urgency of correcting the deficiency or satisfying the need and the impact if not satisfied. The impact should be quantified in

terms of lost training, increased use of aircraft to satisfy training, inability to accomplish mission, etc.

A8.5.3. **(Added-ACC)** Block 10, *CONSTRAINTS/ASSUMPTIONS/PROPOSED SOLUTIONS*: describe the preferred solution(s), known constraints, and assumptions made. Identify risks of the proposed requirement with emphasis on highest risk. If known, state if the modification to satisfy the proposed solution will be contractually procured (formal acquisition process) or provided through government or contractual services (TSSC).

A8.5.4. **(Added-ACC)** Block 12E or Block 17, *Using Command/Lead Command Control No.*, A5T and, as appropriate, other MAJCOM Divisions will assign a number for all AF Forms 1067 that require command validation and/or certification/approval. **(T-2)**

A8.5.5. **(Added-ACC)** Blocks of control numbers listed below are by weapon system. The SCWG chairperson will process the AF Form 1067 for those modifications within the authority of the SCWG. **(T-2)** The "T" suffix is a "TRAINER UNIQUE" identifier.

A8.5.5.1. **(Added-ACC)** A-10. (CY)-001 to 050-T

A8.5.5.2. **(Added-ACC)** E-3. (CY)-051 to 100-T

A8.5.5.3. **(Added-ACC)** E-7. (CY)-101 to 150-T

A8.5.5.4. **(Added-ACC)** F-15. (CY)-151 to 200-T

A8.5.5.5. **(Added-ACC)** F-16. (CY)-201 to 250-T

A8.5.5.6. **(Added-ACC)** F-22. (CY)-251 to 300-T

A8.5.5.7. **(Added-ACC)** F-35. (CY)-301 to 350-T

A8.5.5.8. **(Added-ACC)** RC-135. (CY)-351 to 400-T

A8.5.5.9. **(Added-ACC)** MQ-9. (CY)-401 to 450-T

A8.5.5.10. **(Added-ACC)** HH-60W. (CY)-451 to 500-T

A8.5.5.11. **(Added-ACC)** TACP. (CY)-501 to 550-T

A8.5.5.12. **(Added-ACC)** CRC. (CY)-551 to 600-T

A8.5.5.15. **(Added-ACC)** MC-12. (CY)-701 to 750-T

A8.5.5.16. **(Added-ACC)** HC-130J. (CY)-751 to 800-T

A8.5.5.17. **(Added-ACC)** EC-130H and EA-37B. (CY)-801 to 850-T

A8.5.5.18. **(Added-ACC)** RQ-170. (CY)-851 to 900-T

A8.5.5.19. **(Added-ACC)** T-38. (CY)-901 to 950-T

A8.5.6. **(Added-ACC)** Block 18, *LEAD COMMAND REMARKS*: the LC action officer will coordinate and record HQ staff remarks (assumptions and constraints) in this block, including identification of the MAJCOM OPR/Division and date of coordination. **(T-2)** All validated requirements should include the comment: "Validated as mission essential for the reasons stated in Block 8 and 9." For permanent modifications, include "this is a sustainment modification" or "this is a capability modification."

A8.5.7. **(Added-ACC)** Block 28, *ALSO AFFECTS*: check all applicable blocks.

A8.5.8. **(Added-ACC)** Block 30, *TOTAL COST*: Training System Program office will complete estimated cost.

A8.5.9. **(Added-ACC)** Part V, *LEAD COMMAND (LC) CERTIFICATION, Approval and Funding*: to be completed by the appropriate LC OPR.

A8.6. (Added-ACC) Coordination. A5T and, as appropriate, other MAJCOM divisions coordinate AF Forms 1067 through the appropriate MAJCOM coordination process for validation and approval.